Decipering viral RNA structure with ViennaRNA

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viruses *in silico* lecture series
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universität wien

Part I

RNA structure prediction

with

`ViennaRNA`

Part II

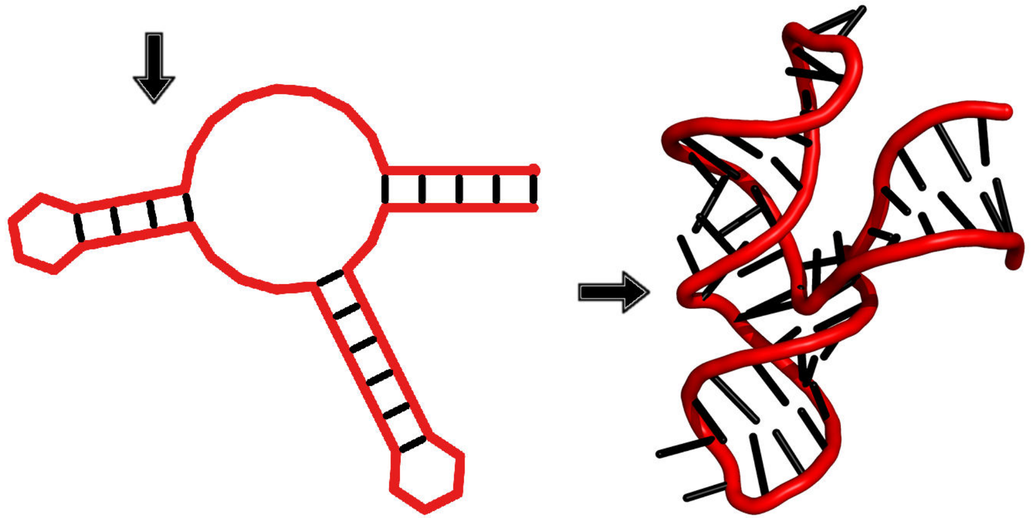
RNA structure conservation

in

Tick-borne encephalitis virus

The RNA folding problem

GCGCUCUGAUGAGGCCGCAAGGCCGAAACUGCCGCAAGGCAGUCAGCGC

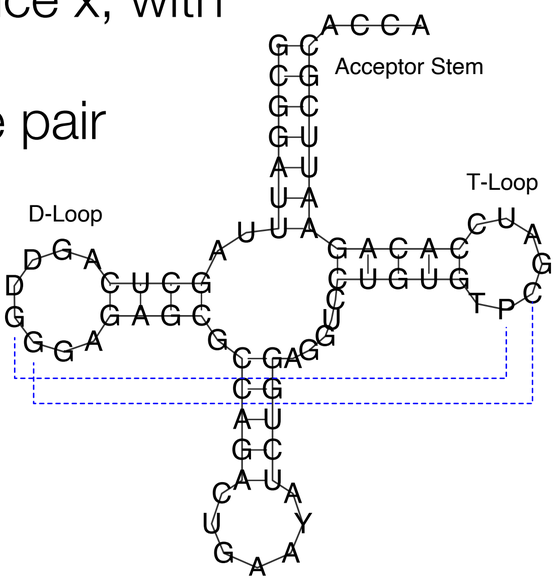


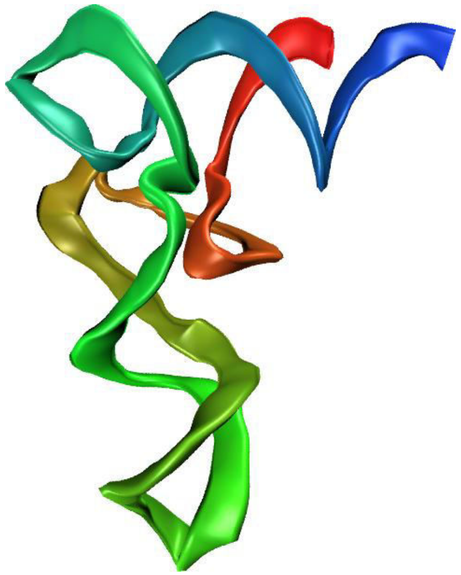
- Hierarchical folding: Secondary structure forms first then helices arrange to form tertiary structure
- Secondary structure captures a majority of stabilising interactions
- Convenient and biologically useful description: Many thermodynamic properties can be predicted
- Starting point for RNA 3D structure prediction
- Tertiary structure prediction needs knowledge of secondary structure

RNA secondary structures

A secondary structure is a list of base pairs (i,j) on a sequence x, with

- Any nucleotide (sequence position) can form at most one pair
- If (i,j) is a pair then $x_i x_j \in \{GC,CG,AU,UA,GU,UG\}$
- If (i,j) is a base pair, then $j - i > 3$
- No pseudo-knots: No pairs (i,j) and (k,l) with $i < k < j < l$

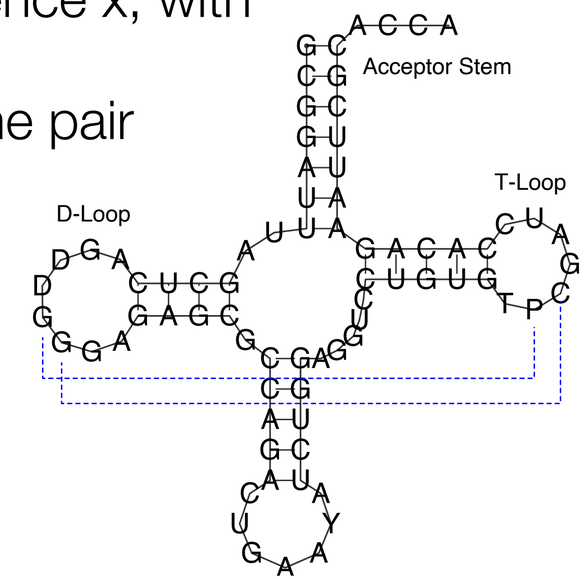


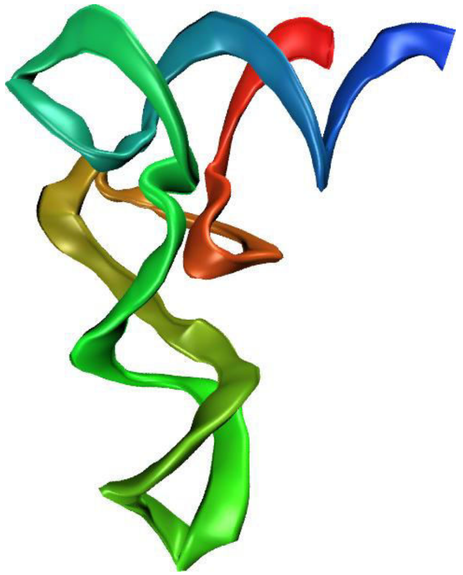


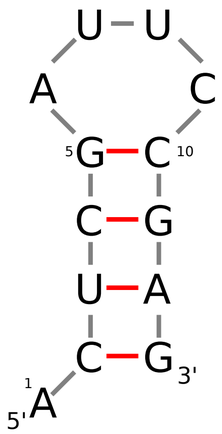
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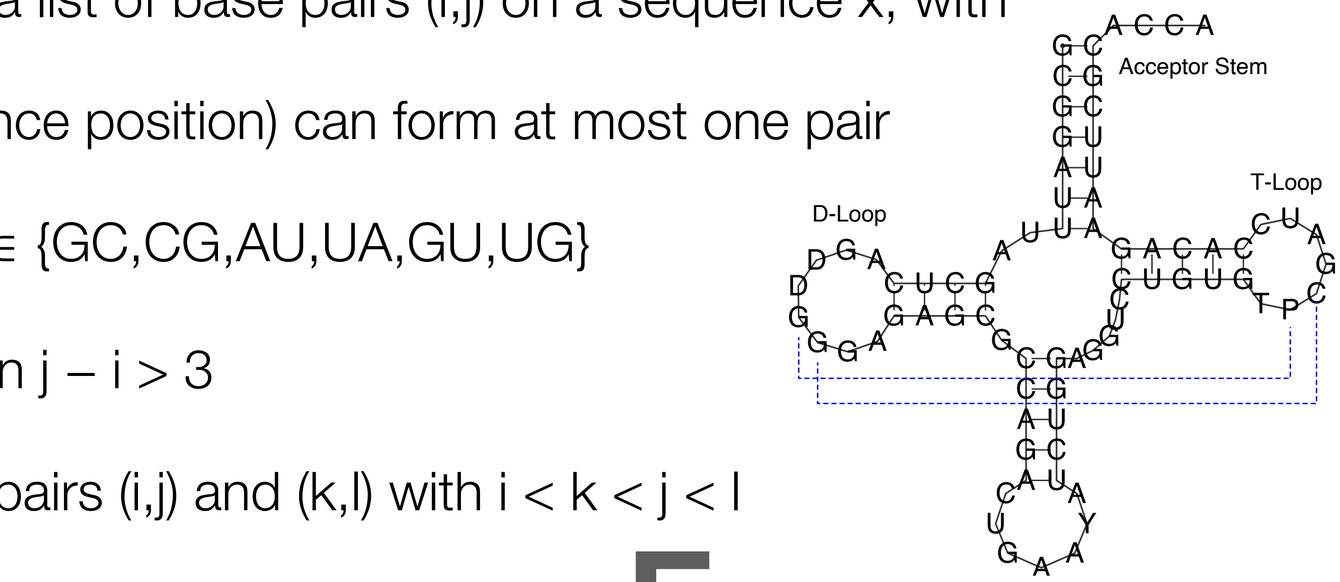


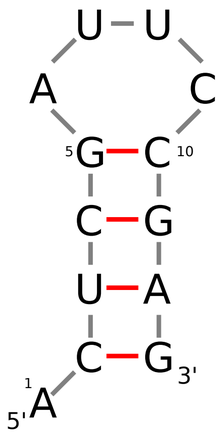

```
A C U C G A U U C C G A G
. ( ( ( ( . . . . ) ) ) )
```

RNA secondary structures

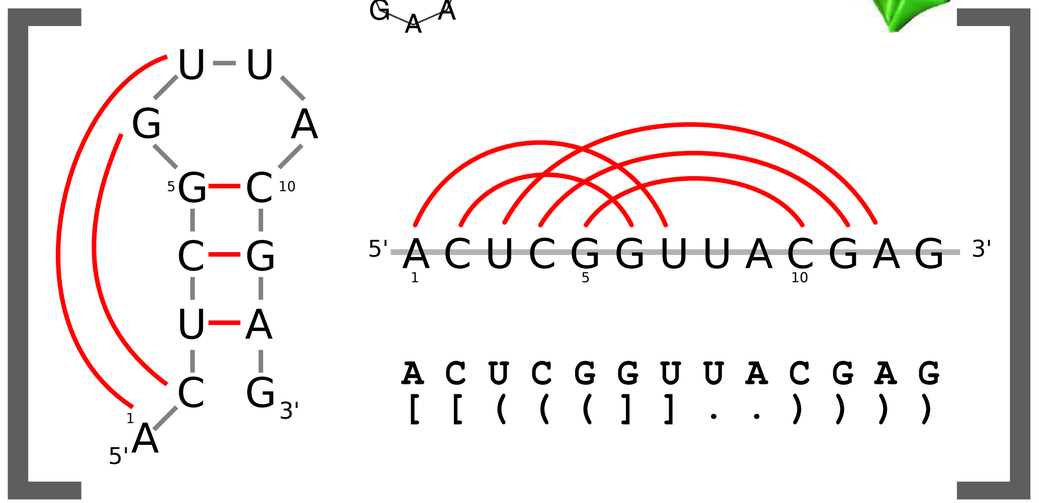
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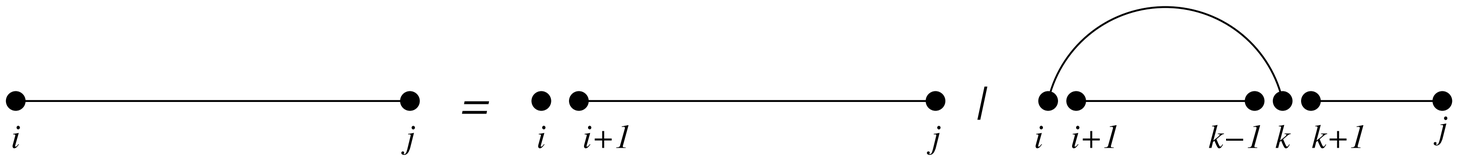




Solving the RNA folding problem

Toy model for RNA folding: assign energies to base pairs $\varepsilon(x, y)$

Easily solved by Dynamic Programming: recursive computation with tabulation of intermediate results



$$E_{ij} = \min_{i<k\leq j} \left\{ E_{i+1,j} \; ; \quad \left(E_{i+1,k-1} + E_{k+1,j} + \varepsilon(x_i, x_k) \right) \right\}$$

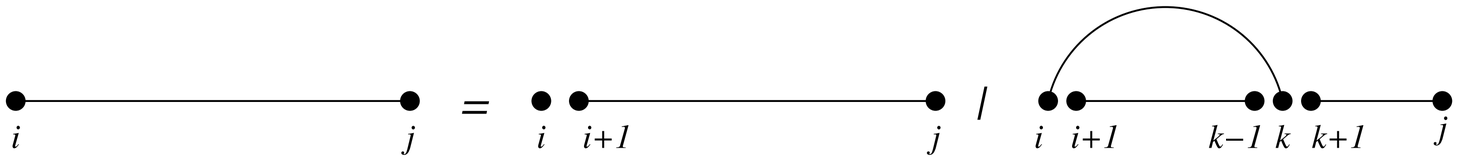
- E_{1n} is the best possible energy for our sequence
- Backtracing through the E table yields the corresponding structure
- The algorithms requires $\mathcal{O}(n^2)$ memory and $\mathcal{O}(n^3)$ CPU time

In practice this toy model is not good enough !
We need loop-dependent energies for serious predictions

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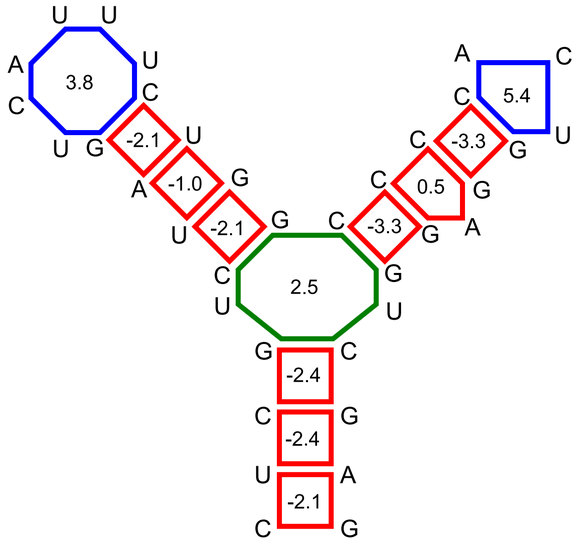


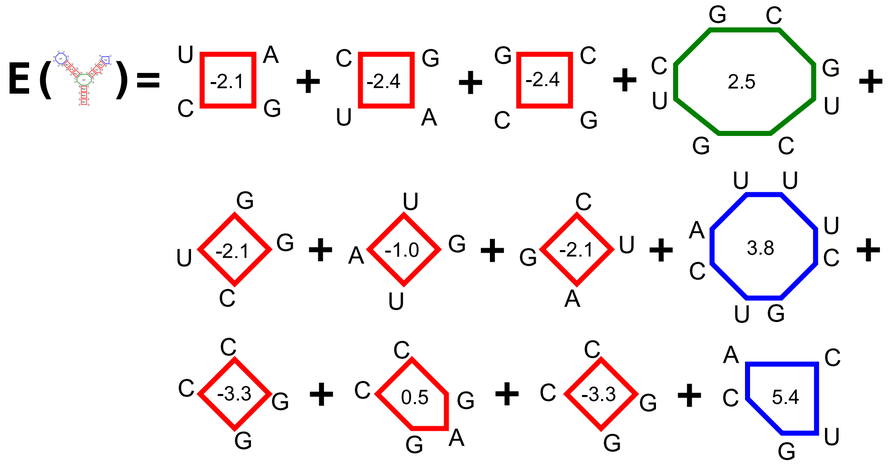
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Nearest neighbor energy model





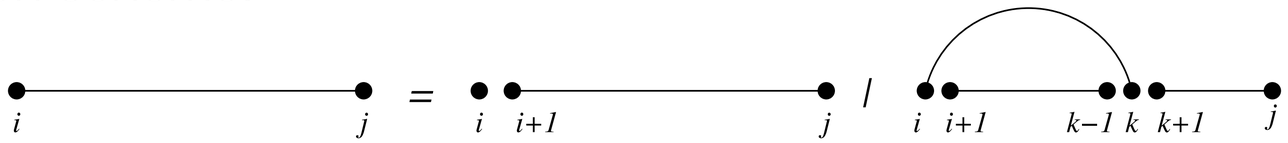
- Secondary structure *s* is uniquely decomposed into loops *L*
- Contributions of a base pair only depends on neighbouring pairs
- Each loop *L* is assigned a free energy contribution E_L [1]

$$E(s) \approx \sum_{L \in s} E_L$$

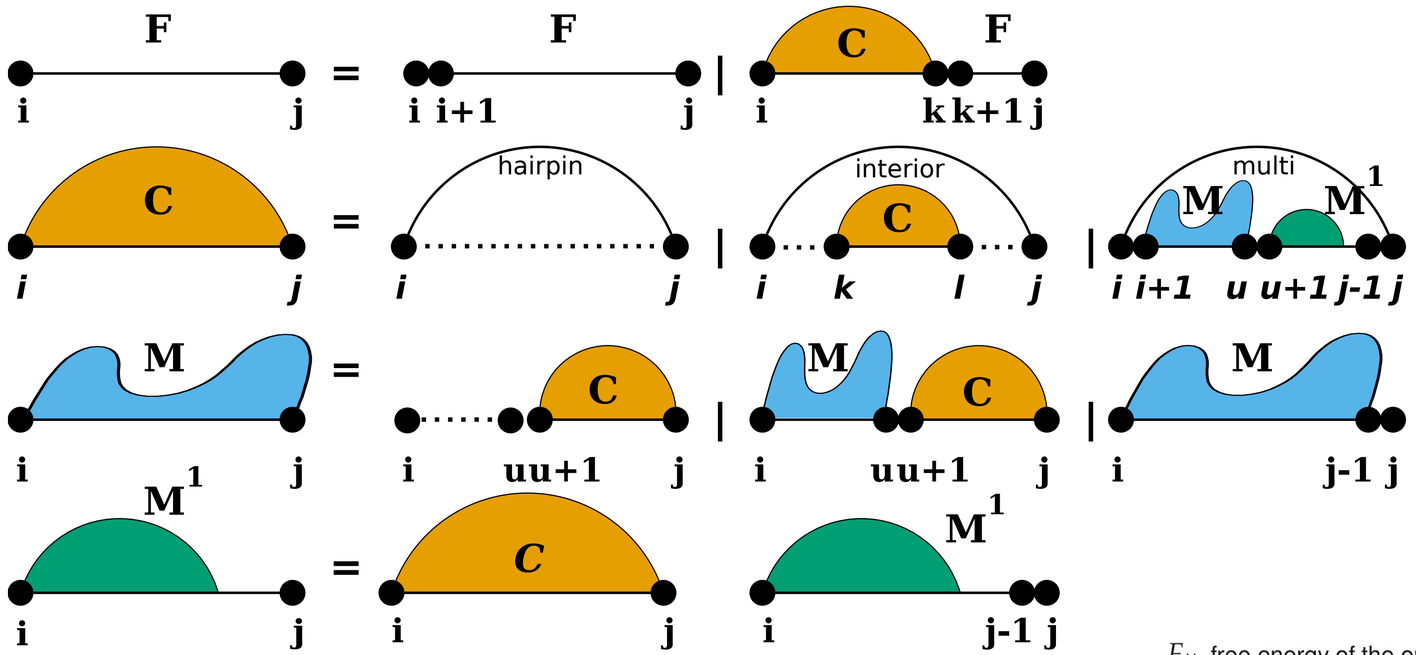
[1] *D. Turner et.al. (2009) NAR 38, D280-D282*

RNA secondary structure prediction

Nussinov decomposition scheme:



Full decomposition scheme:



F_{ij} free energy of the optimal substructure on the subsequence $x[i..j]$.

C_{ij} optimal free energy on $x[i..j]$, where (i, j) pair.

M_{ij} $x[i..j]$ is part of a multiloop and contains at least one pair.

M^1_{ij} same as M_{ij} but contains exactly one component closed by (i, h).

RNA secondary structure prediction

- Minimum free energy (MFE)
- Suboptimal secondary structures
- Partition function Z

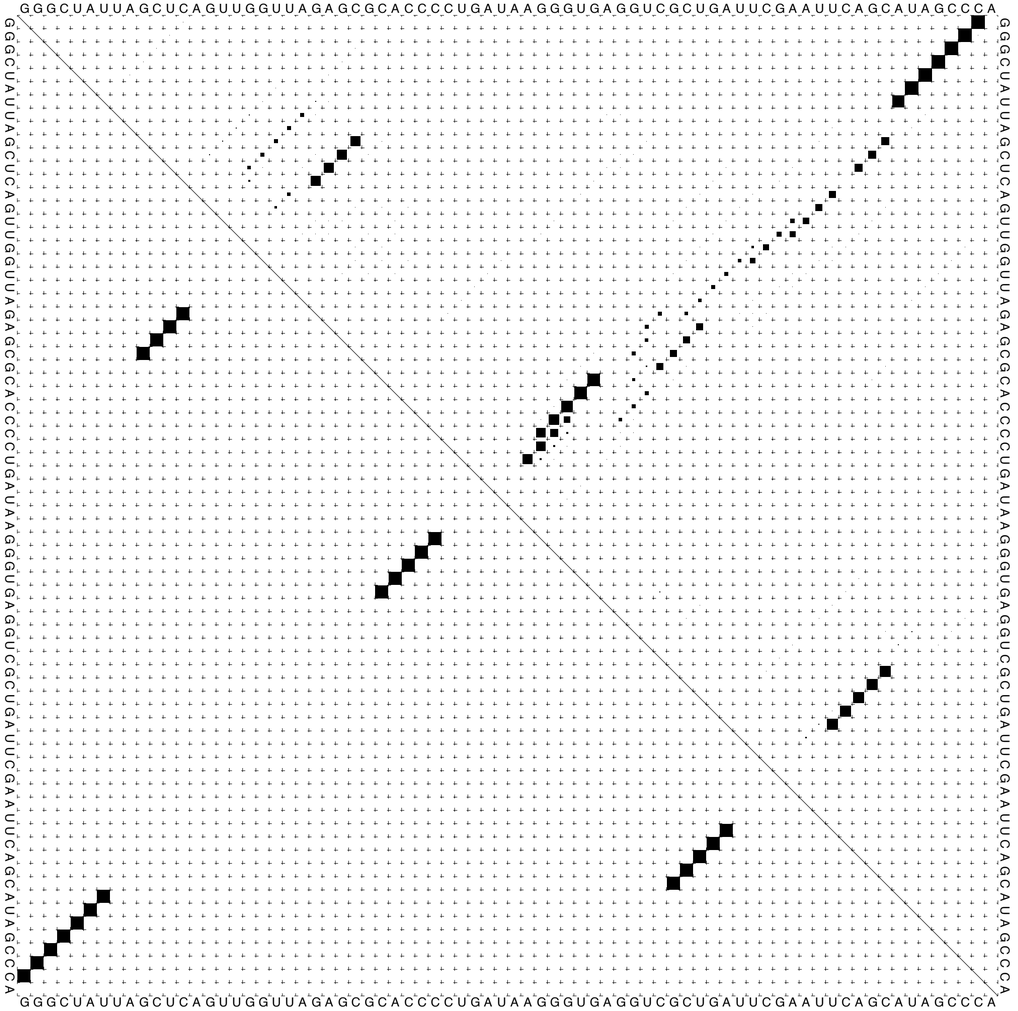
$$p(s) = e^{-\beta E(s)} \quad \text{with} \quad \beta = \frac{1}{kT} \longrightarrow Z = \sum_{s \in \Omega} e^{-\beta E(s)}$$

- Probability of feature $\mathcal{F}$

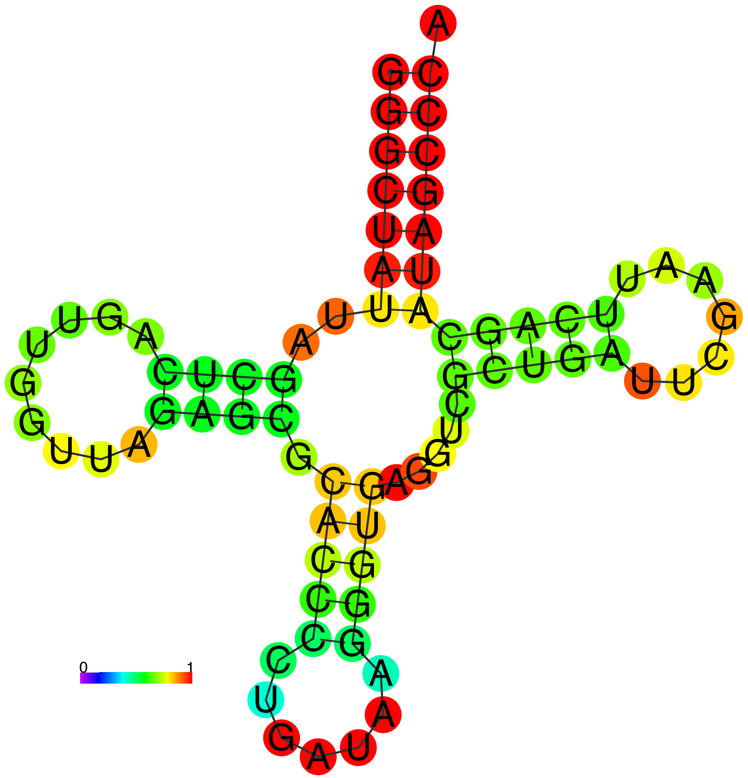
$$p(\mathcal{F}) = \frac{Z_{\mathcal{F}}}{Z} \quad \text{with} \quad Z_{\mathcal{F}} = \sum_{s|F \in s} e^{-\beta E(s)}$$

- Base pair probabilities
- Stochastic backtracking (Boltzmann smapling)
- Global/local reliability measures

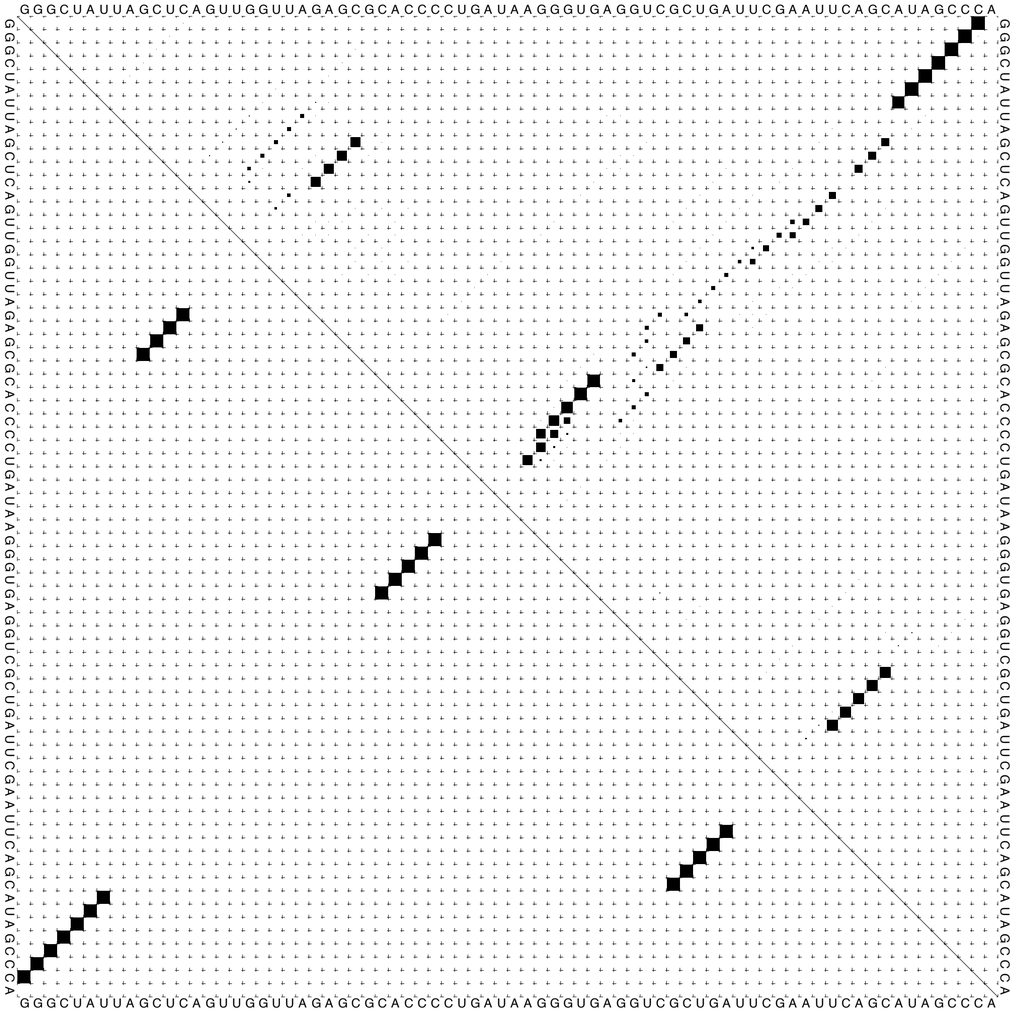
Base pair probabilities



Base pair probability dot plot

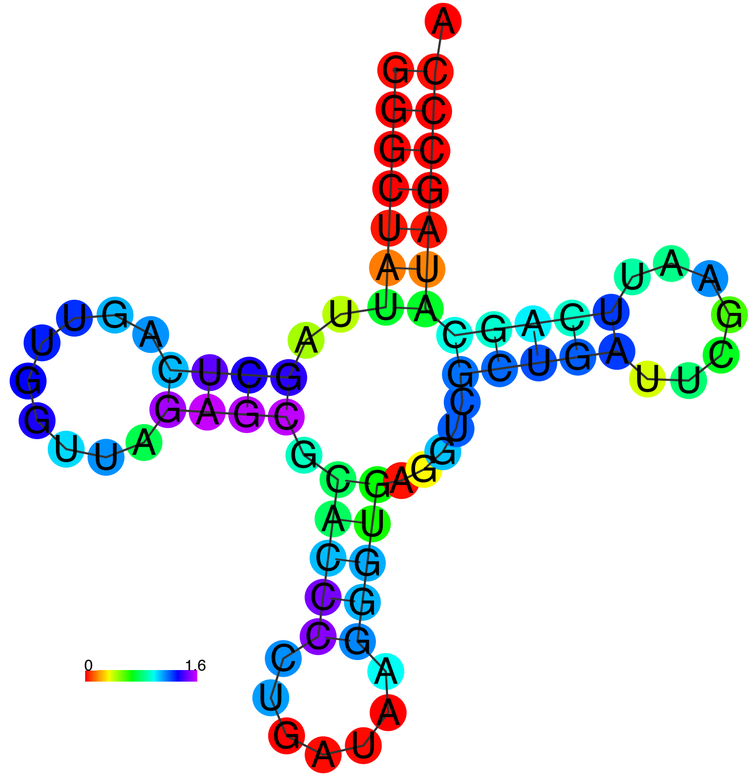


Well-definedness: Positional entropy



Local reliability:

Which parts of the prediction can we trust?



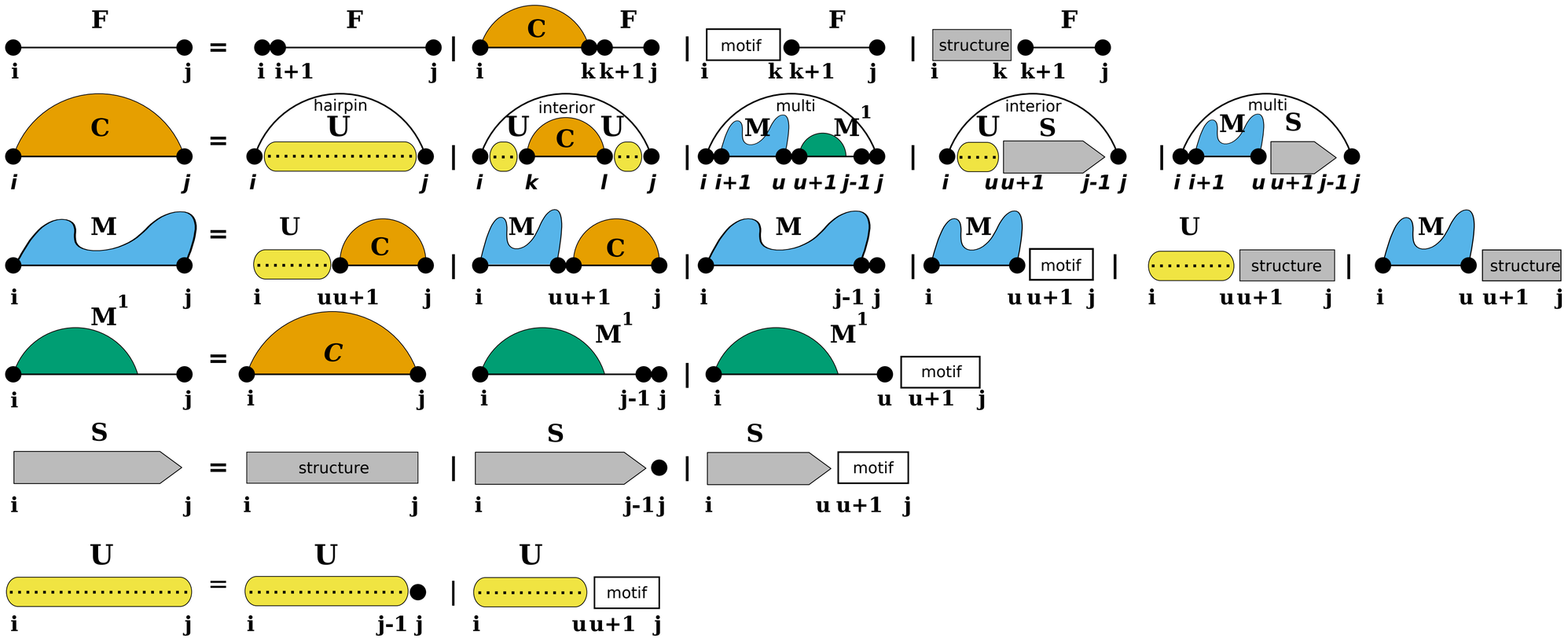
Positional (‘Shannon’) entropy is computed from pair probabilities

$$\mathrm{PE}(i) = -\sum_k p_{ik} \ln p_{ik}$$

RNA secondary structure prediction

The ViennaRNA Package enables the decomposition schema (folding grammar) to be:

- Constrained in terms of candidate space (hard constraints)
- Constrained in terms of candidate evaluation (soft constraints)
- Extended by self-enclosed structured/unstructured domains



Applications to constraint folding

Candidate space (hard constraints):

- Force | prohibit nucleotides to pair, e.g. within a pore
- Force | prohibit base pair formation, e.g. cross-linked pairs

Applications to constraint folding

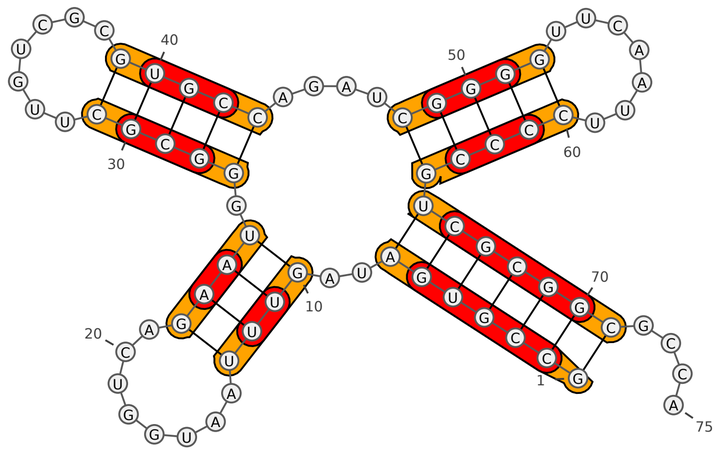
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Candidate evaluation (soft constraints):

- Include chemical probing data to guide structure prediction, e.g. SHAPE reactivity data

Deigan [2]



$$\Delta G(i) = m * ln(\text{reactivity}[i] + 1) + b$$

- Reactivity probes backbone flexibility (unpairedness)
- No nucleobase bias
- Convert reactivity to pseudo-energy for structure prediction

[2] *Deigan et.al. (2009) PNAS 106, 97-102*

Applications to constraint folding

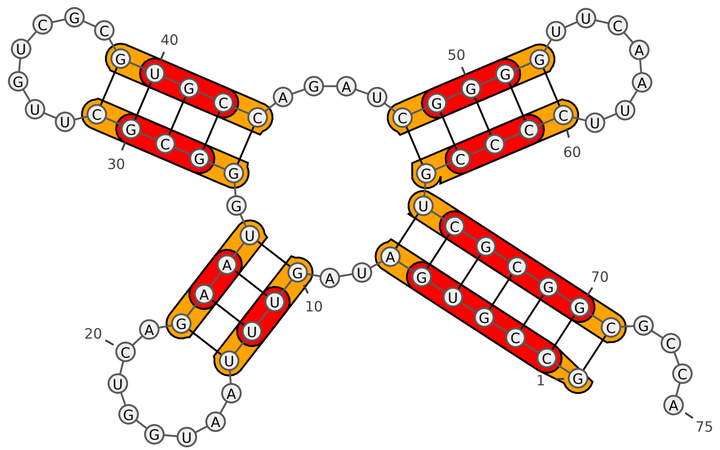
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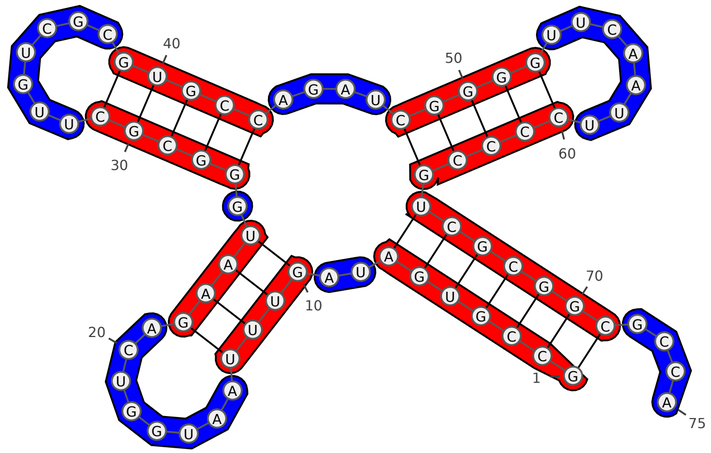
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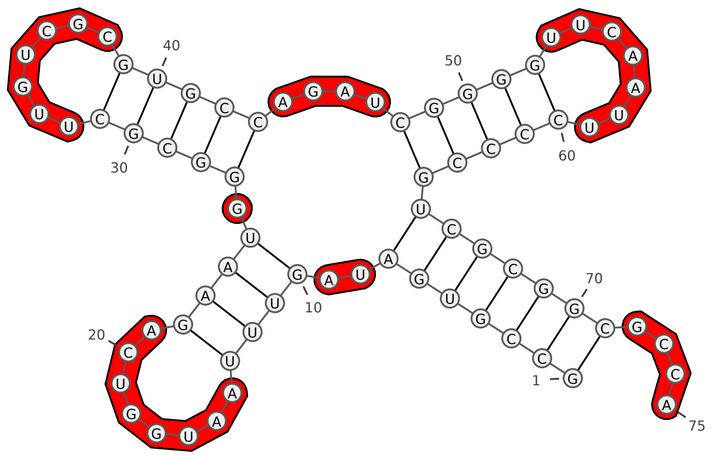
$$\Delta G(i) = m * ln(\text{reactivity}[i] + 1) + b$$

Zarringhalam [3]



$$\Delta\ G(x,i) = \beta\ * |x - q_i|$$
$$x \in [0(\text{unpaired}), 1(\text{paired})]$$

Washietl [4]



$$F(\vec{\epsilon}) = \sum_{i=1}^{n} \frac{\epsilon_i^2}{\tau^2} + \sum_{i=1}^{n} \frac{(p_i(\vec{\epsilon}) - q_i)^2}{\sigma^2} \rightarrow \min$$

[2] *Deigan et.al. (2009) PNAS 106, 97-102*

[3] *Zaringhalam et.al. (2012) PLoS One 7, e45160*

[4] *Washietl et.al. (2012) NAR 40, 4261-72*

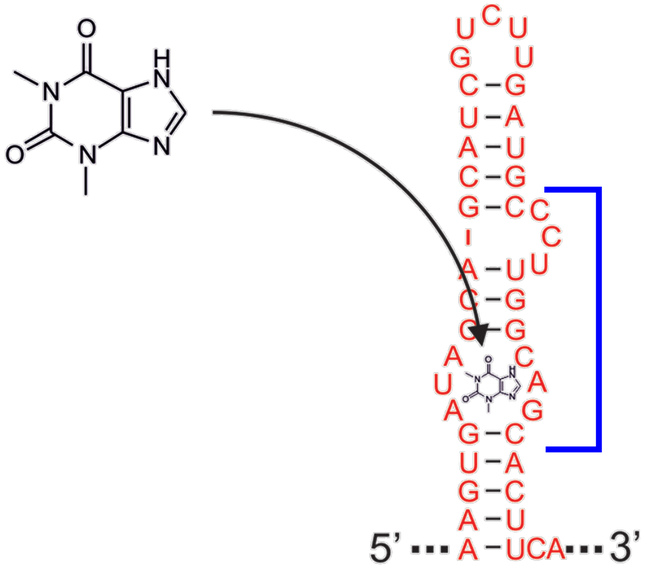
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Candidate evaluation (soft constraints):

- Include chemical probing data to guide structure prediction, e.g. SHAPE reactivity data
- Model (more or less) complex RNA-ligand interactions for interior loops and hairpin loops




```
$ RNAfold -p --motif="GAUACCAG&CCCUUGGCAGC,(...((((&)...)))...),-9.22" --verbose
>theo-P-IS10
read ligand motif: GAUACCAG&CCCUUGGCAGC, (...((((&)...)))...), -9.220000
GGUGAUACCAGAUUUCGCGAAAAAUCCCUUGGCAGCACCUCGCACAUCUUGUUGUCUGAUUAUUGAUUUUUCGCGAAACCAUUUGAUCAUAUGACAAGAUUGAG
((((...(((((((.....)))))...)))...))))...((.((((((((((.((((((.((.((((....)))).)).)..)))))).)))))))))).. (-33.82)
specified motif detected in MFE structure: (4,36) (11,26)
((((...(((((((.....)))))...)))...))))...,,.((((((((((.((((((.((.((((....)))).)).)..)))))).)))))))))).. [-35.14]
((((...(((((((.....)))))...)))...))))......((((((((((.((((((.((.((((....)))).)).)..)))))).))))))))).... {-24.20 d=4.35}
specified motif detected in centroid structure: (4,36) (11,26)
 frequency of mfe structure in ensemble 0.116952; ensemble diversity 6.71
```

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Structured domains:

- G-Quadruplexes

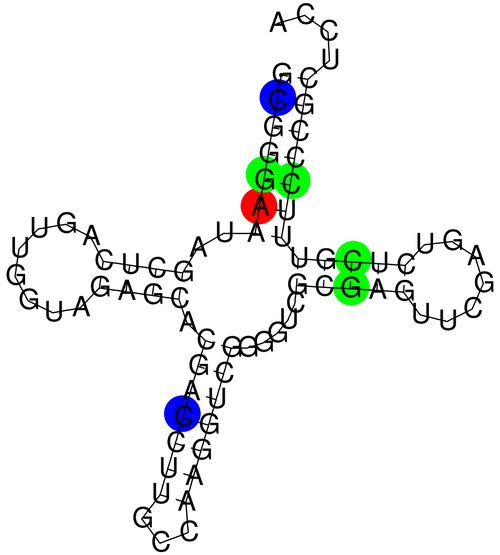
Unstructured domains:

- (IUPAC) motif based default ligand binding for MFE/partition function of single RNA sequences
- Model (more or less) complex RNA-ligand interactions for interior loops and hairpin loops

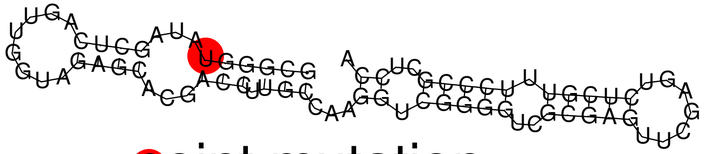
RNA-RNA interaction

- **`RNAduplex`**
  Simplified duplex interaction (intermolecular bp): MFE, suboptimal structures
- **`RNAcofold`**
  Sequence concatenation (inter/intramol. bp): MFE, suboptimal, partition function, concentration dependency
- **`RNAup`**
  Two-step process (inter/intramolecular bp; kissing hairpins): MFE, suboptimal structures
- **`RNAplex`**
  Fast heuristic for genome-wide interaction prediction: MFE, suboptimal structures
- **`RNAmultifold`**
  Concatenation approach for multiple strands: MFE, suboptimal, partition function, concentration dependency

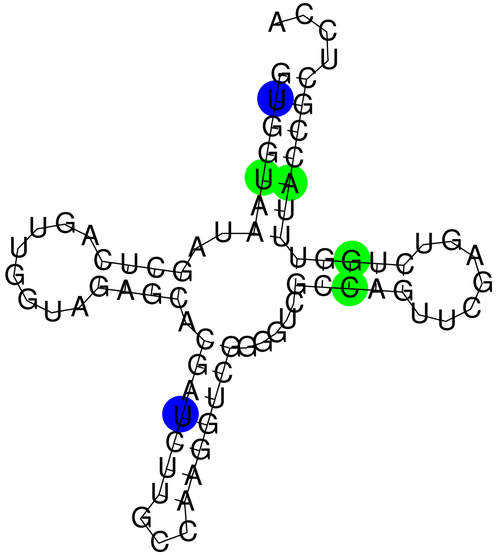
Mutations affect RNA structure



original structure

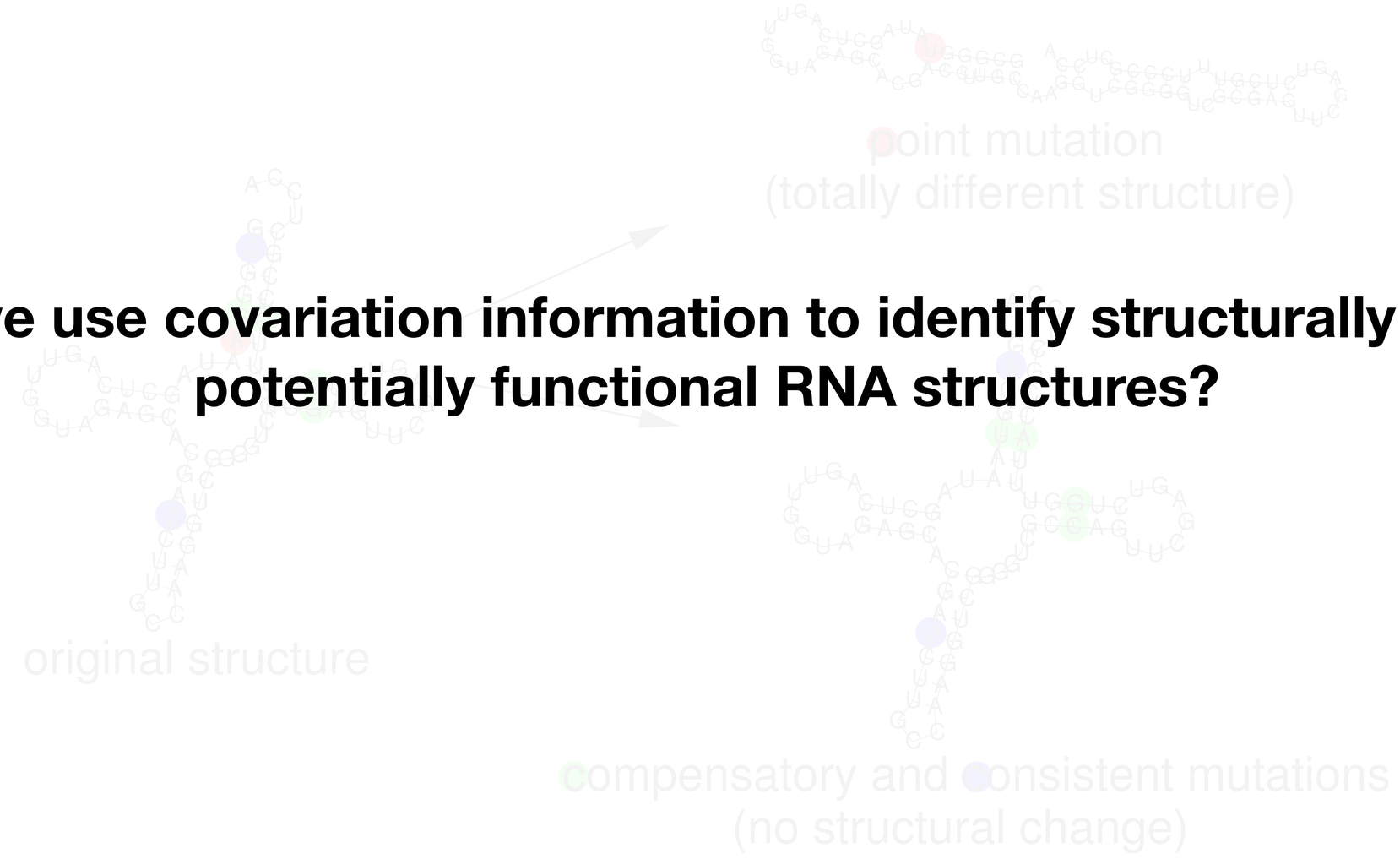


point mutation
(totally different structure)



compensatory and consistent mutations
(no structural change)

Mutations affect RNA structure



How can we use covariation information to identify structurally conserved, potentially functional RNA structures?

Consensus structures: Alignment folding

Combine covariance analysis and folding into one DP algorithm

- Apply conventional folding algorithm to alignment
- Use a modified energy function that includes covariance score

$$E_c(A, \Psi) = \sum_k E(A_k, \Psi) + cv \cdot \sum_{(i,j) \in \Psi} B_{ij}$$

- Can be used for all variants: MFE, partition function, …
- Efficient: $\mathcal{O}(N \cdot n^2 + n^3)$ CPU and $\mathcal{O}(n^2)$ memory for alignment length n and N sequences
- Same results as RNAfold for single sequences

Consensus structures: Alignment folding

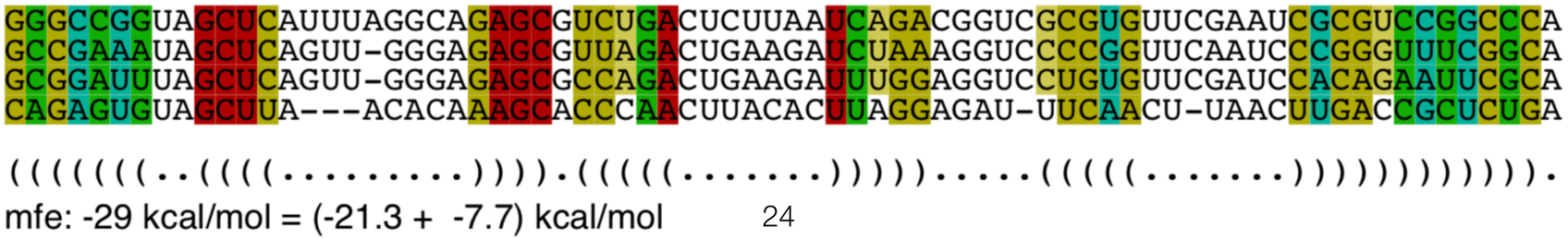
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RNAalifold

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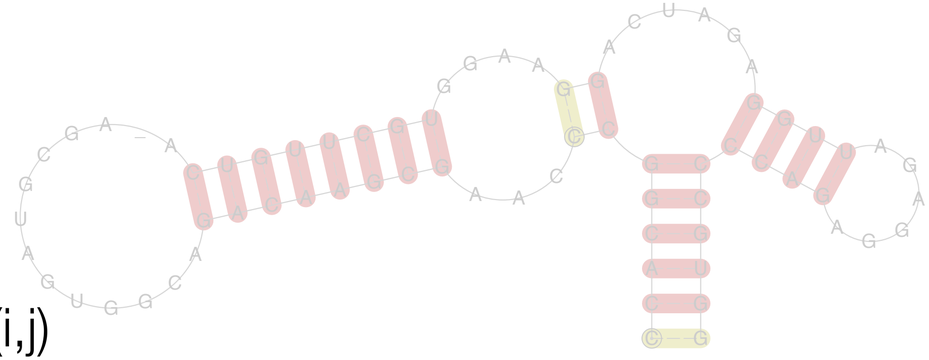
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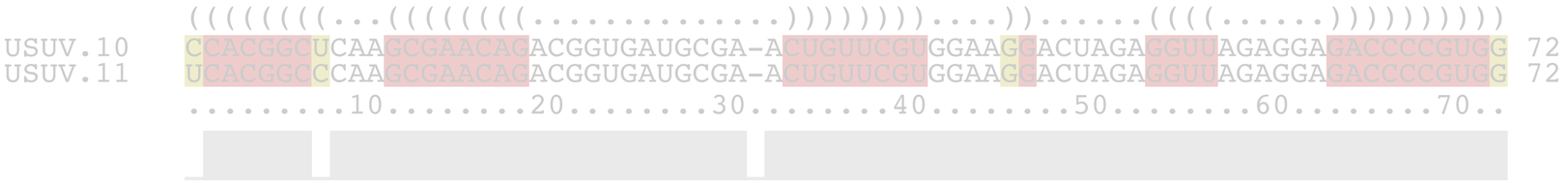
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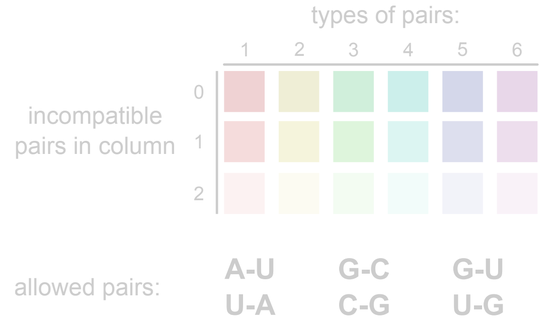


RNA covariation as evolutionary trait

- High mutation rate in RNA viruses due to error-prone RdRP
- For base pair (i,j): GC/CG/AU/UA/GU/UG
- Consistent mutation: different standard combinations
- Compensatory mutation: both positions are mutated
- Presence of both strongly supports predicted base pair (i,j)

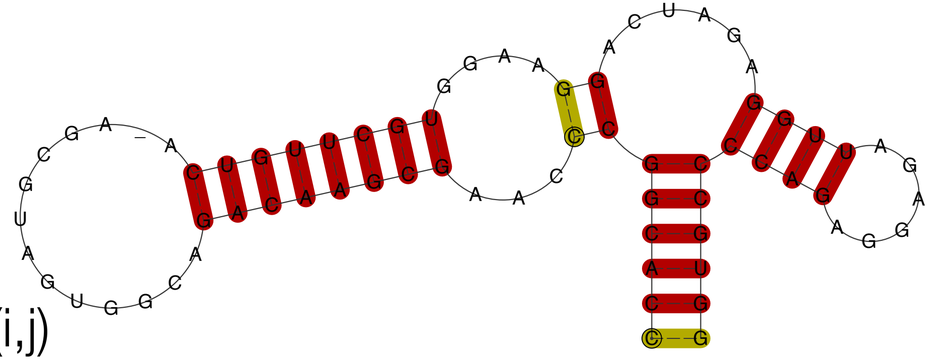


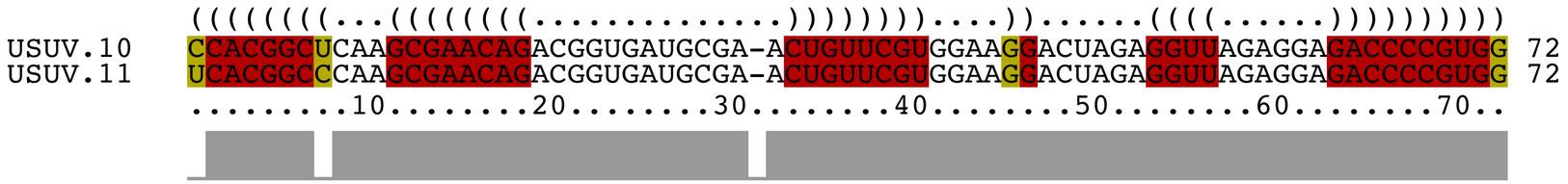


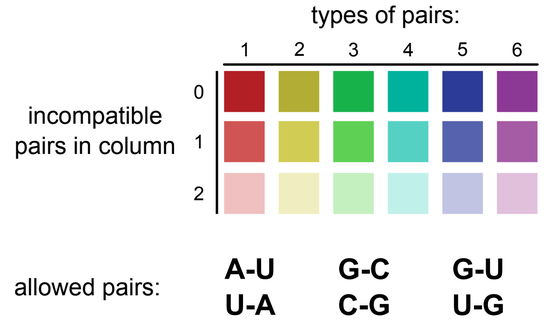


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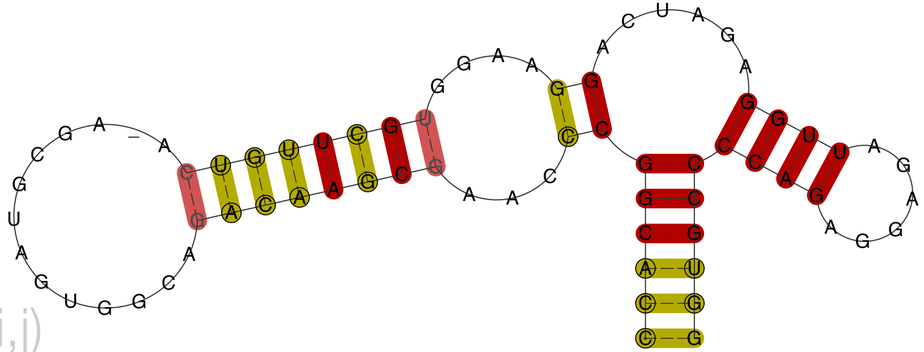


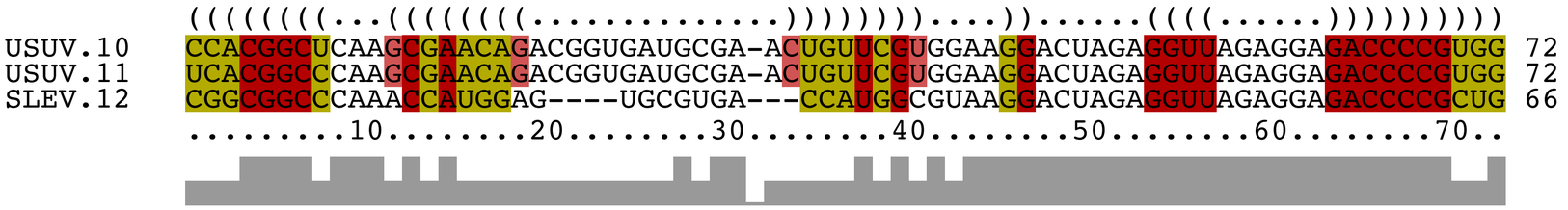


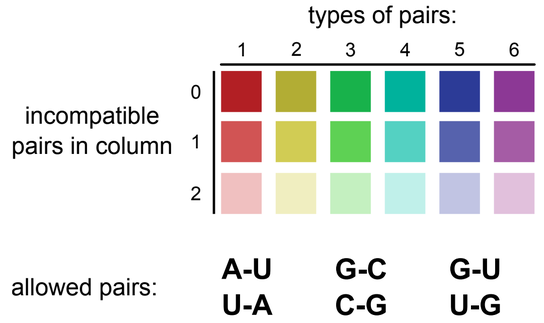


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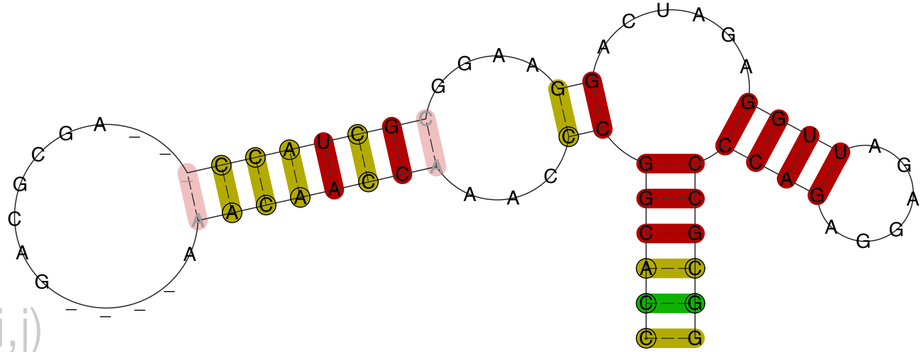


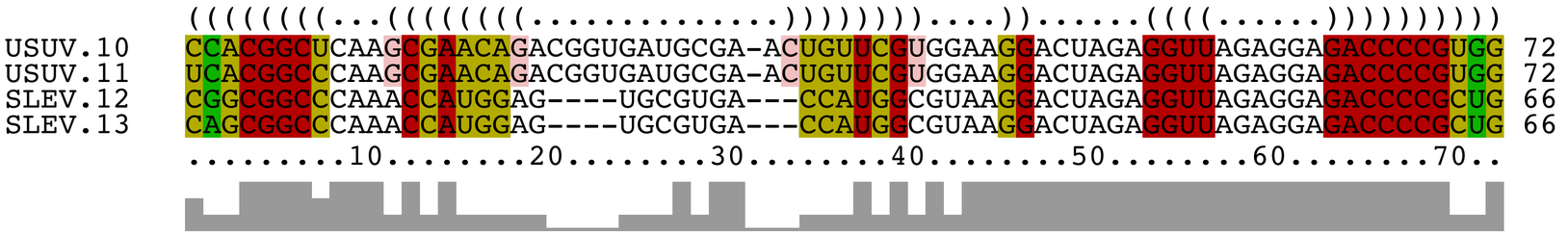


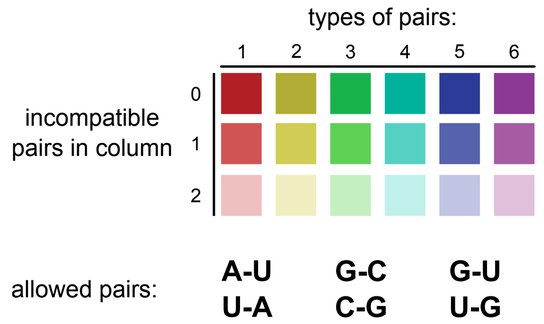


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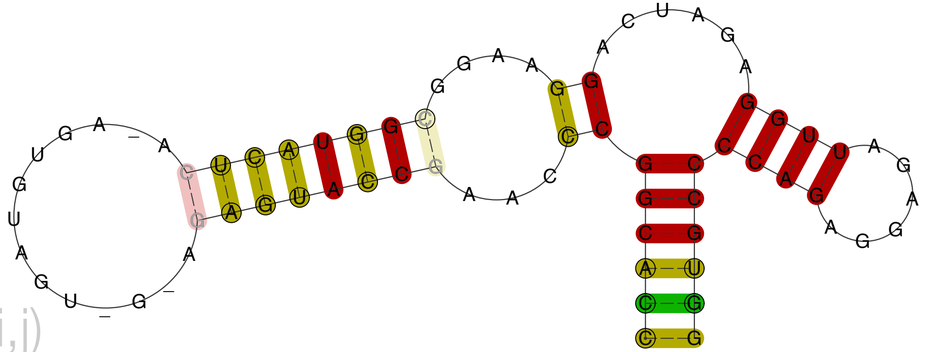


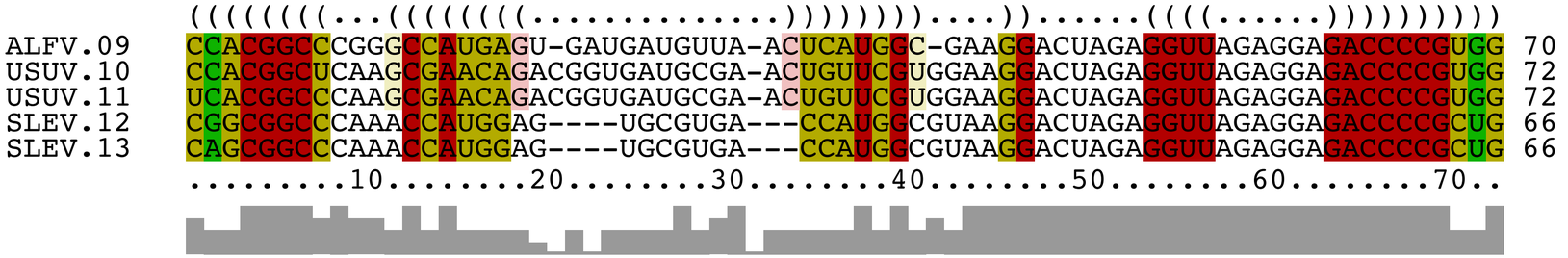


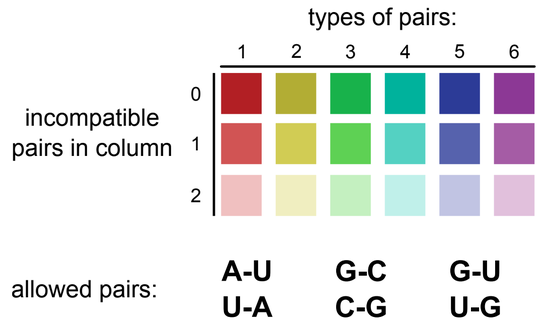


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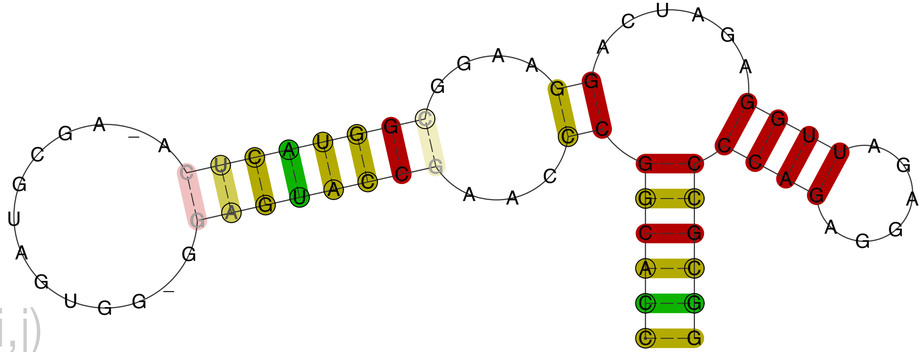


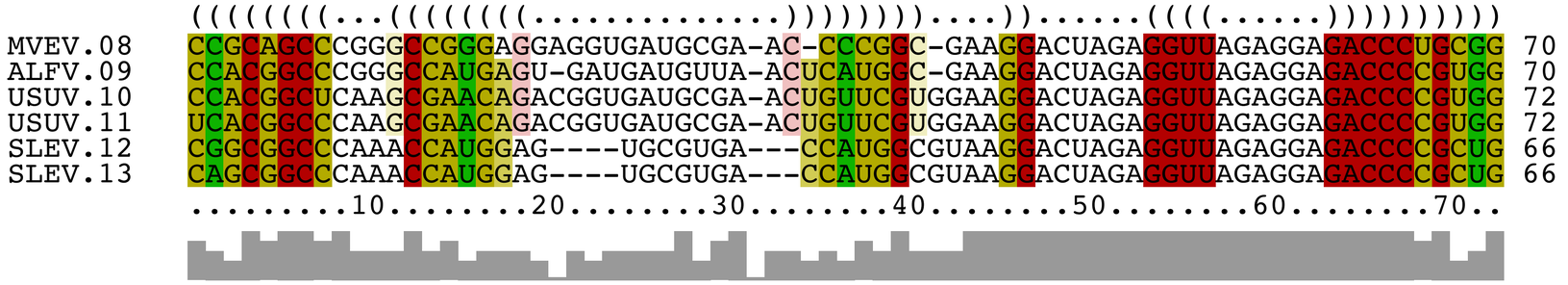


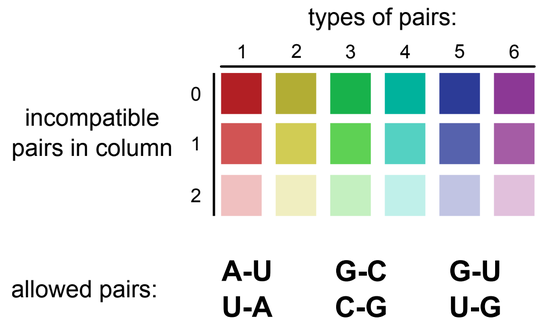


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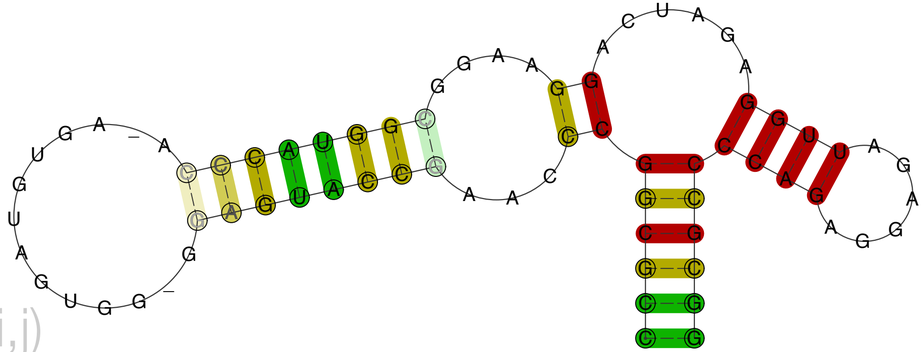


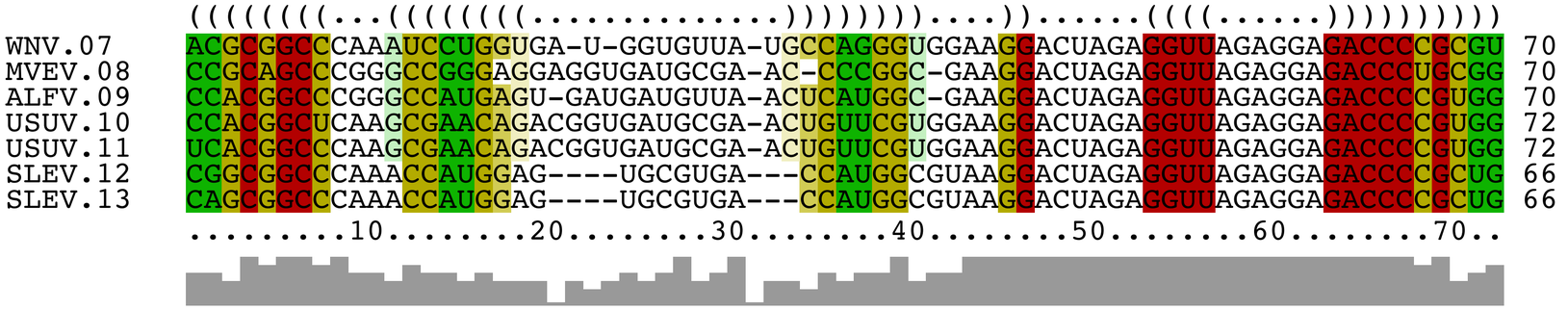


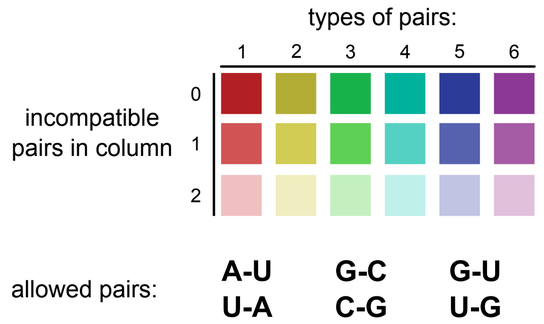


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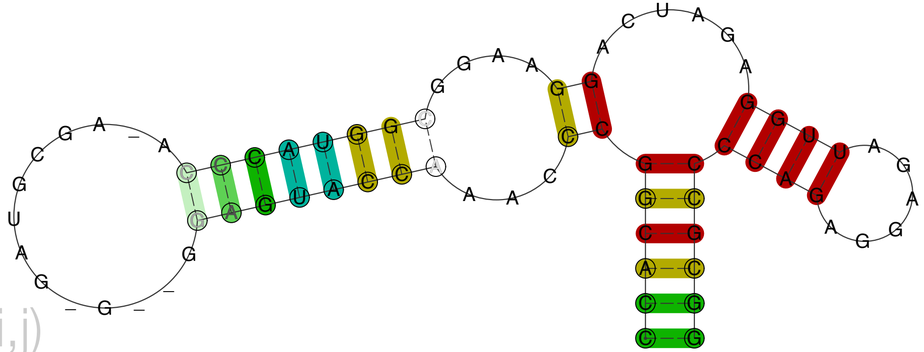


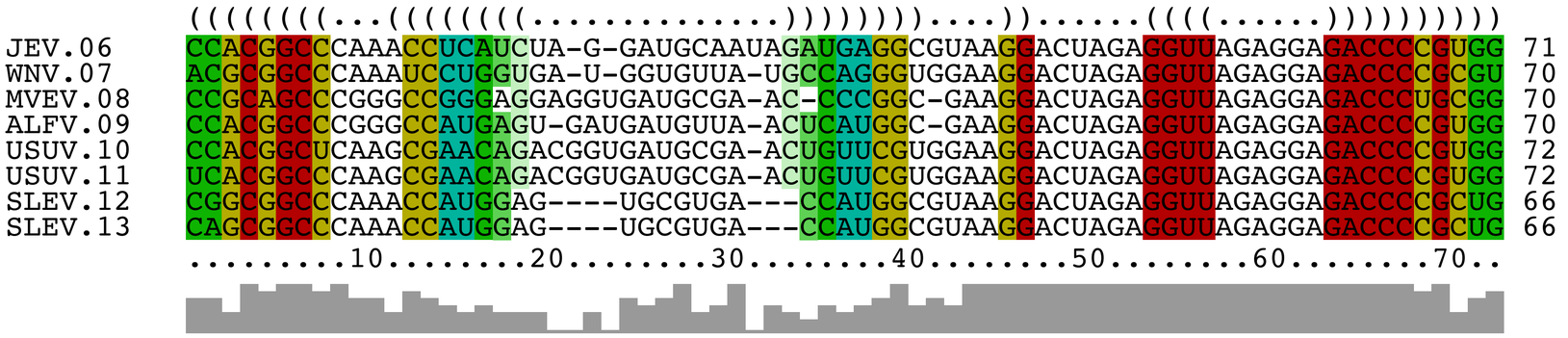


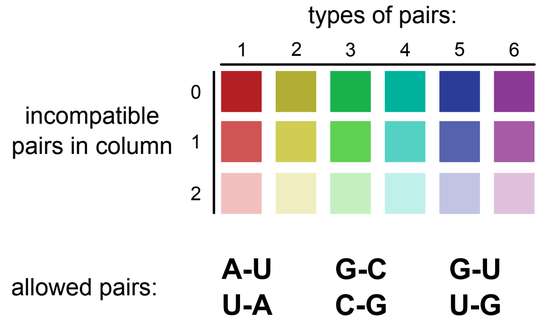


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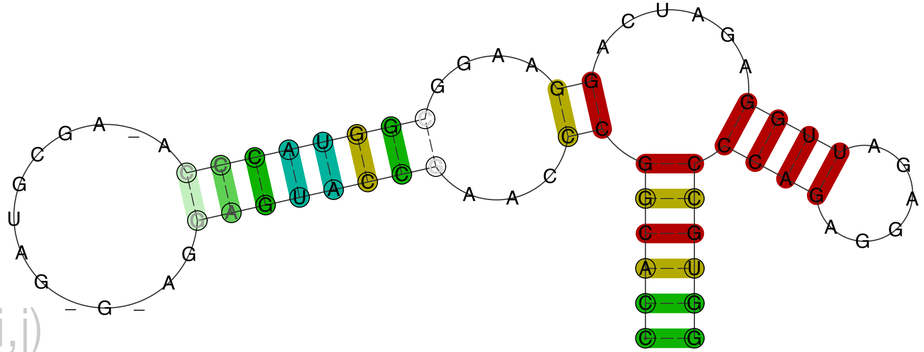


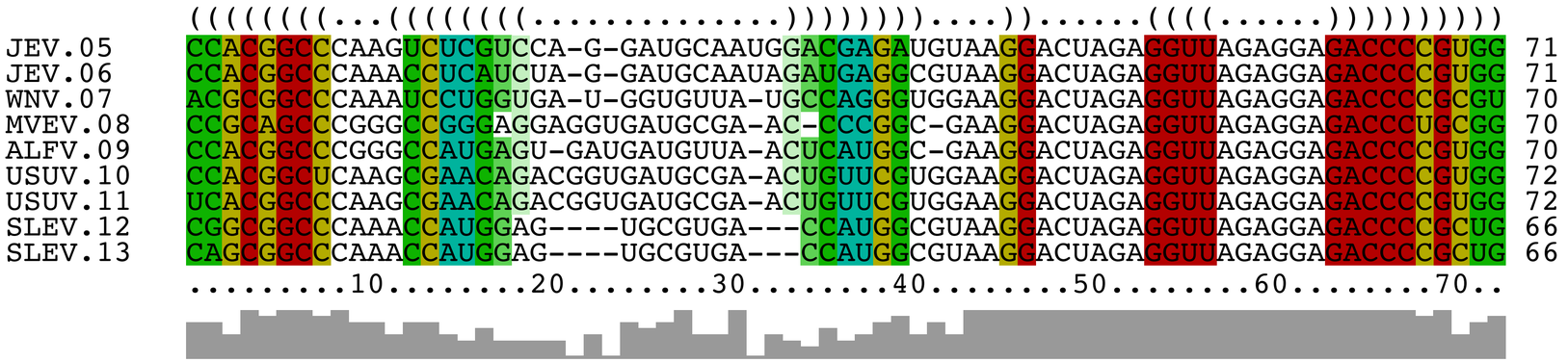


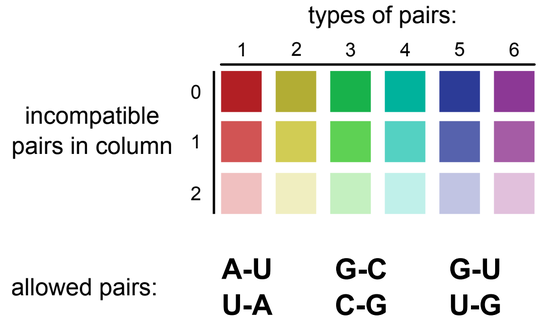


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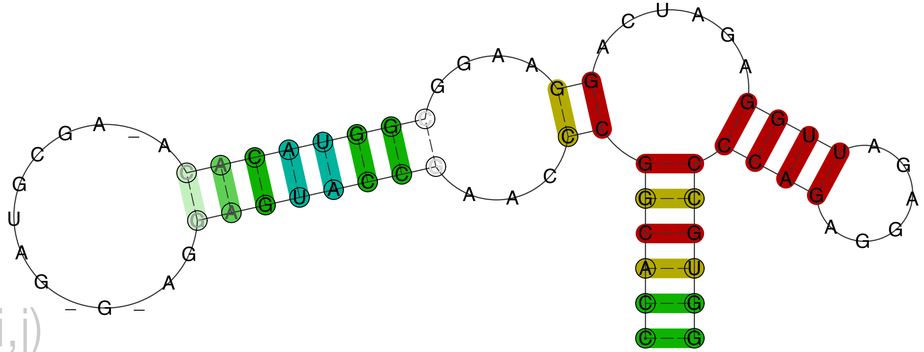


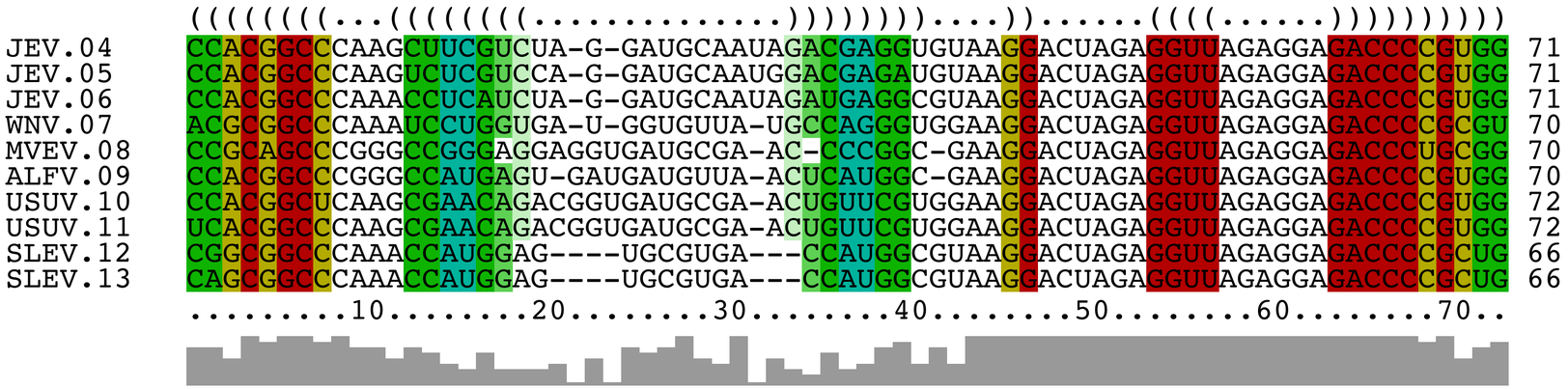


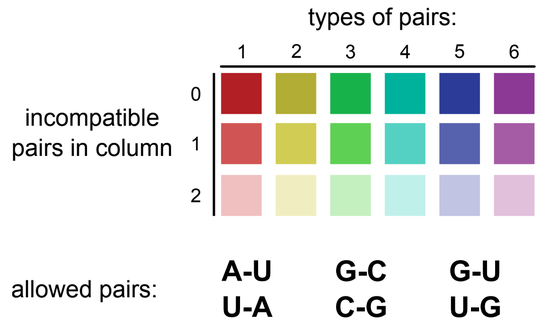


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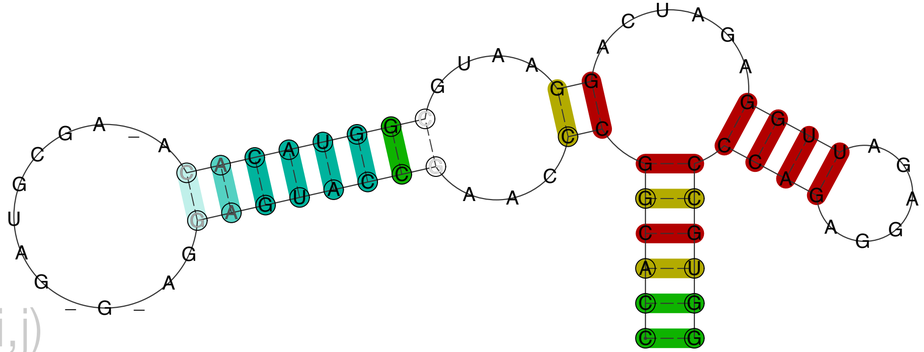


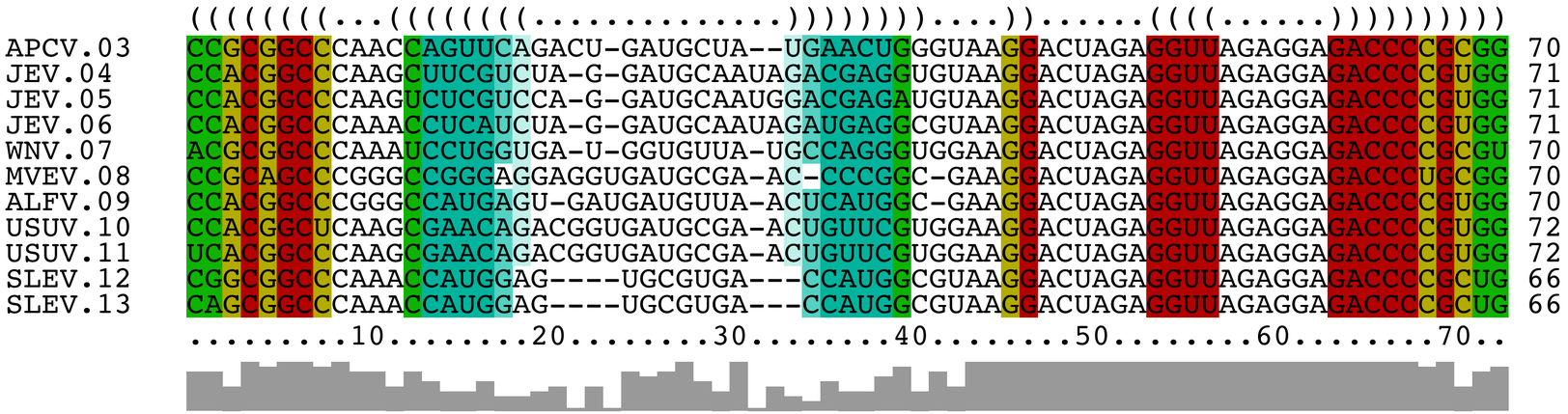


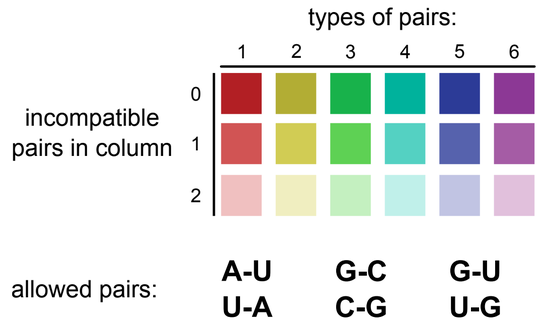


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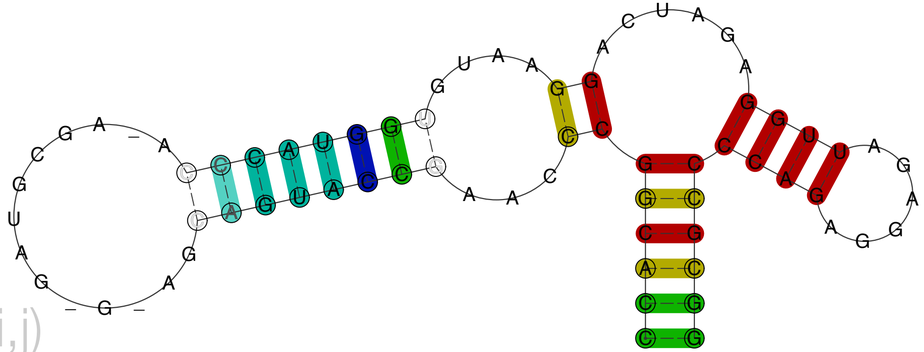


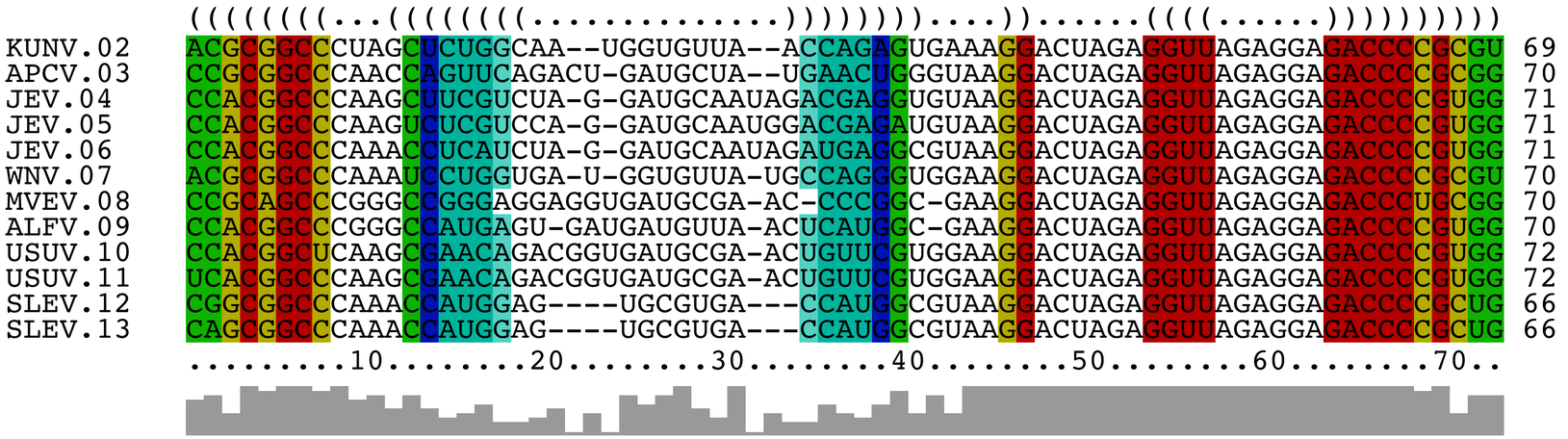


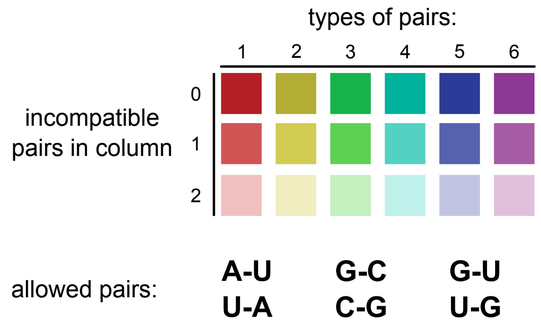


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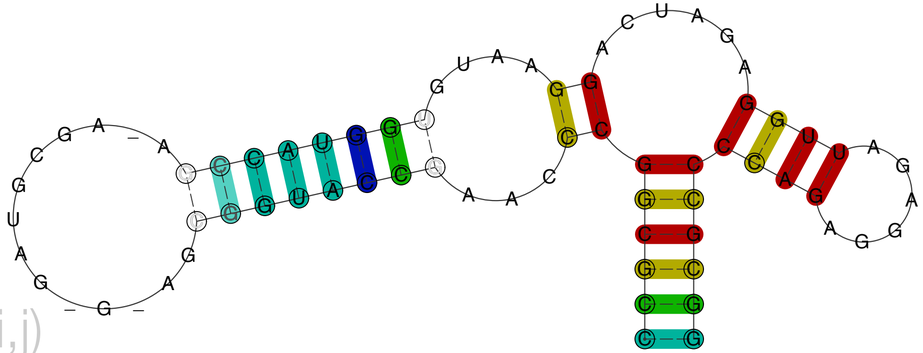


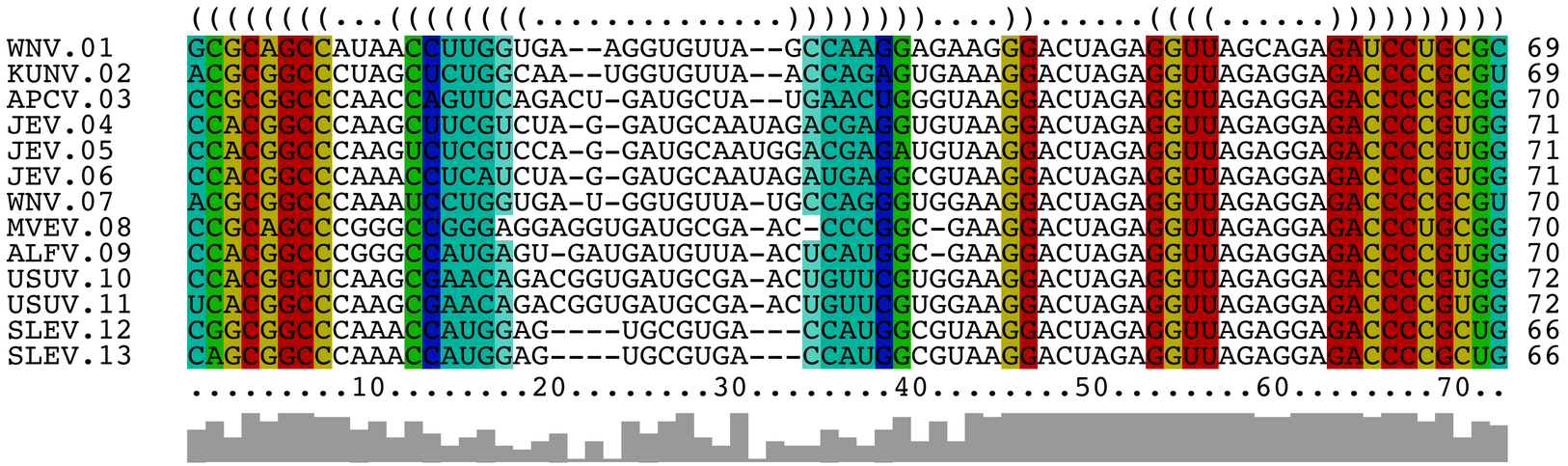


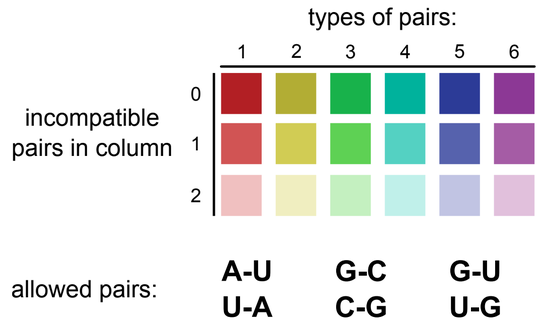


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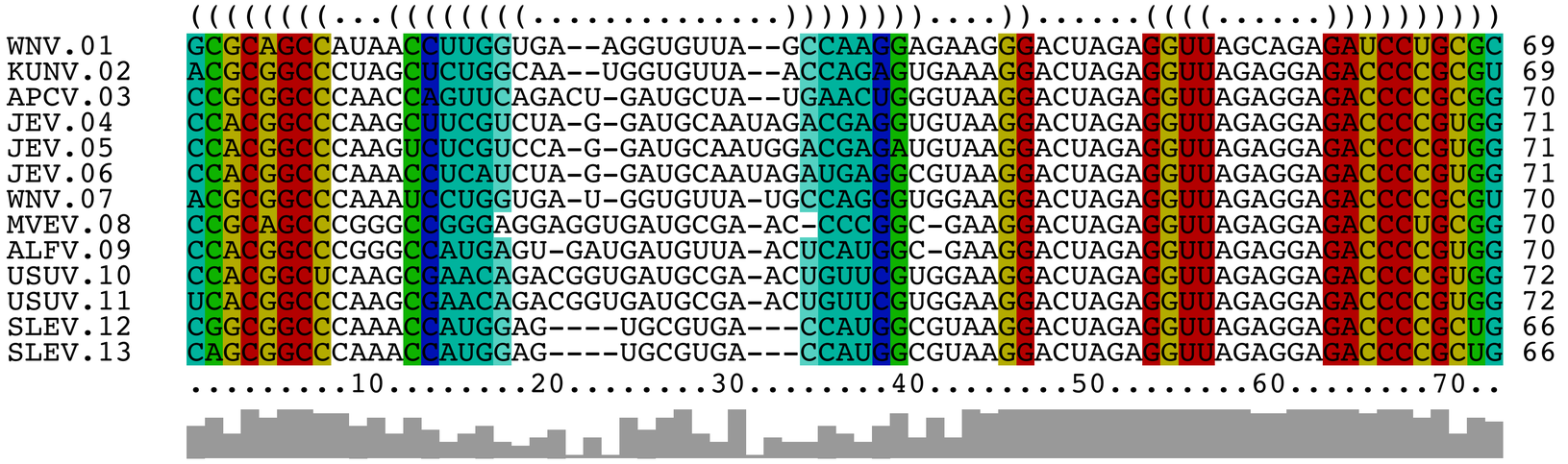


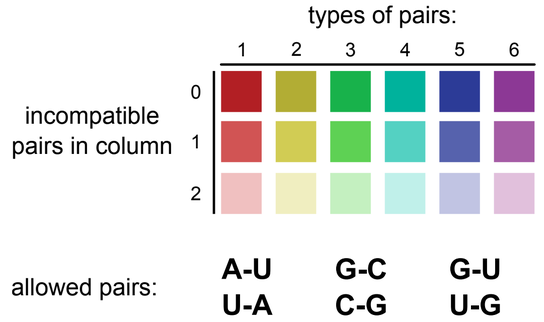


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We use structural RNA alignments and Covariance Models to find conserved elements





The Vienna RNA Package

- Efficient implementations for single, multiple sequence and MSA folding
- Global and local structure prediction
- Prediction of consensus structures
- Hard and soft constraints
- Complete suboptimal folding and energy landscape analysis
- Analysis of folding kinetics / co-transcriptional folding
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www.tbi.univie.ac.at/RNA

Part I

RNA structure prediction

with

ViennaRNA

Part II

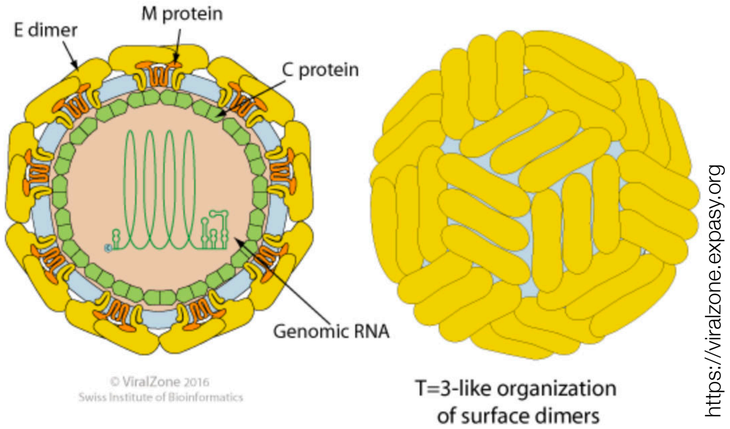
RNA structure conservation

in

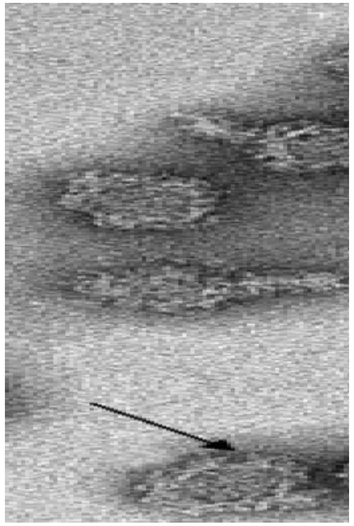
Tick-borne encephalitis virus

Tick-borne encephalitis virus (TBEV)

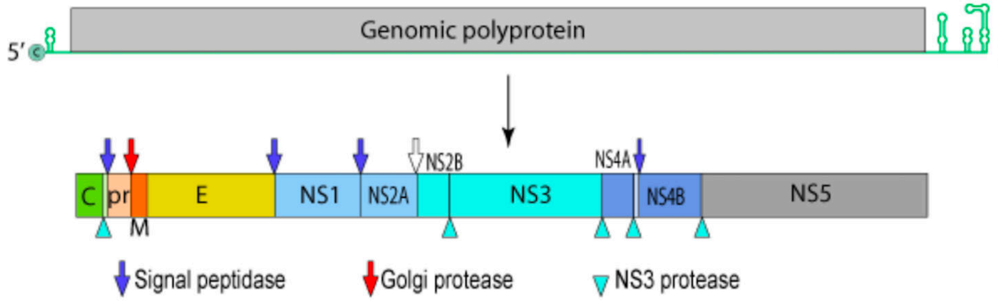
- Zoonotic RNA virus
  (family *Flaviviridae* / genus Flavivirus*)*
- Endemic in forested parts of Eurasia and Japan
- Transmitted between haematophagous ticks and vertebrate hosts (reservoir hosts: small wild-living animals, rodents)
- Thousands of human infections/year. Majority is asymptomatic or subclinical, but TBEV has neurotropic potential and can cause meningitis, encephalitis or haemorrhagic syndrome
- Vaccines available



https://viralzone.expasy.org



https://en.wikipedia.org

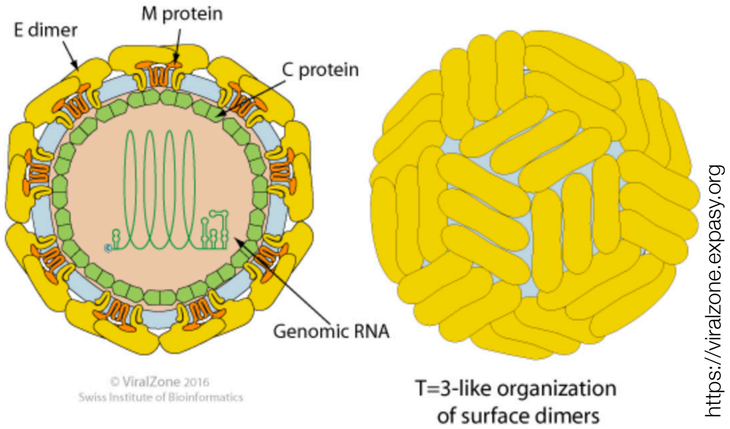


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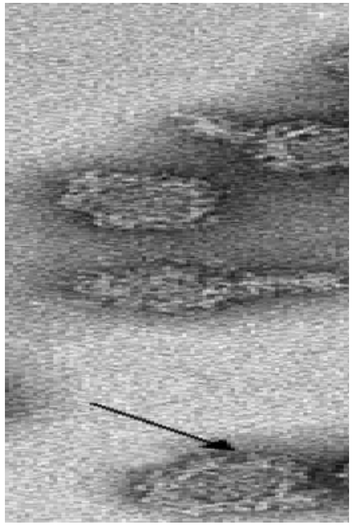
- Non-segmented, single-stranded, (+)-sense RNA genomes of ~11kB length
- Capped, non-polyadenylated
- Encodes a single ORF, flanked by highly structures 5’UTR and 3’UTR

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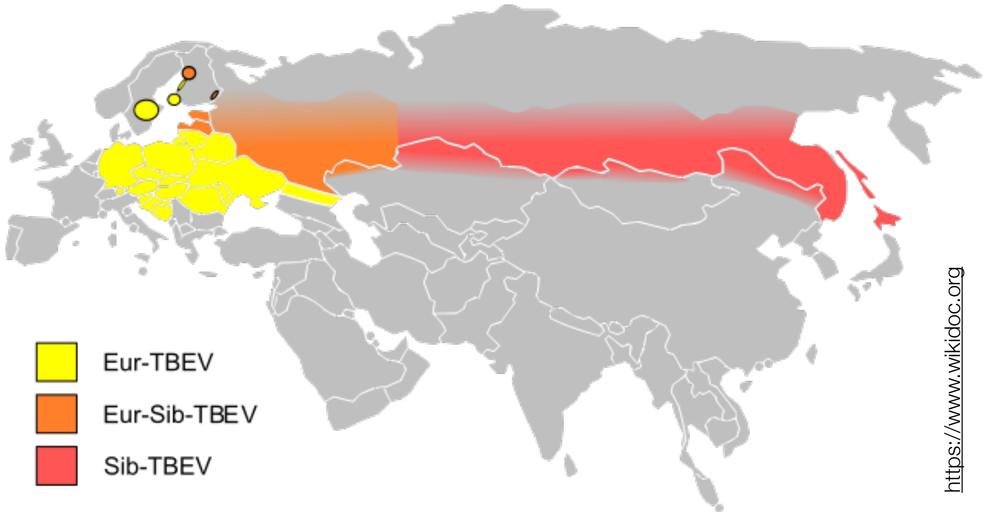
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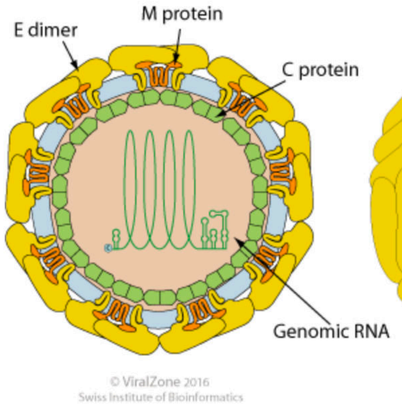
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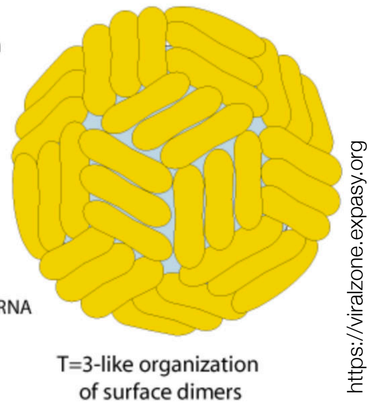


https://www.wikidoc.org

Tick-borne encephalitis virus (TBEV)

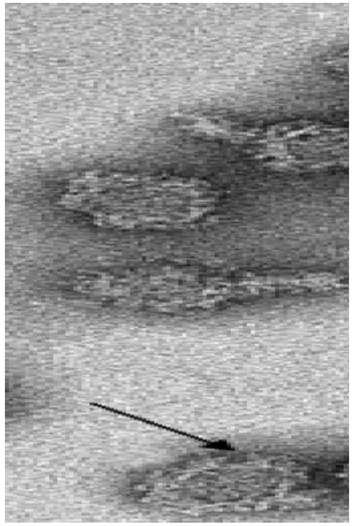
- Zoonotic RNA virus
  (family *Flaviviridae* / genus Flavivirus*)*
- Endemic in forested parts of Eurasia and Japan
- Transmitted between haematophagous ticks and vertebrate hosts (reservoir hosts: small wild-living animals, rodents)
- Thousands of human infections/year. Majority is asymptomatic or subclinical, but TBEV has neurotropic potential and can cause meningitis, encephalitis or haemorrhagic syndrome
- Vaccines available



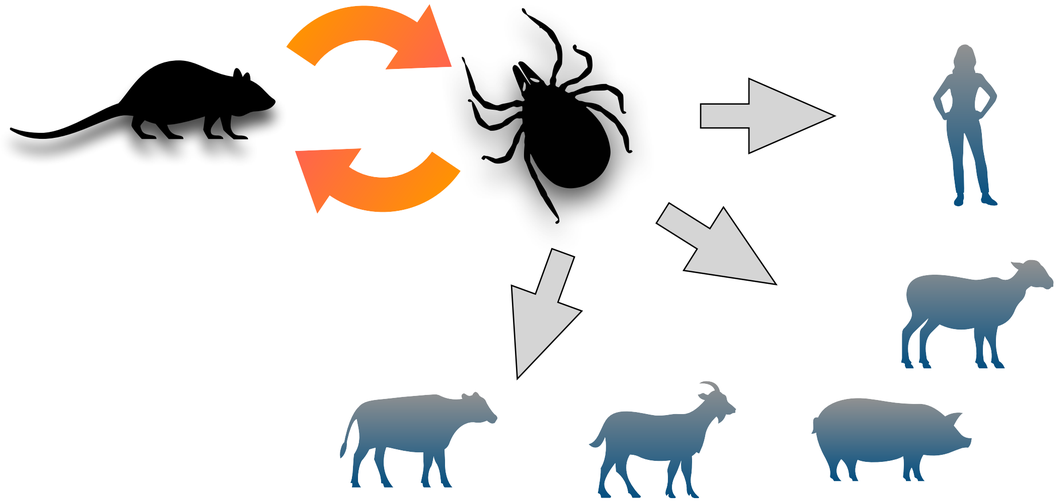


T=3-like organization of surface dimers

https://viralzone.expasy.org

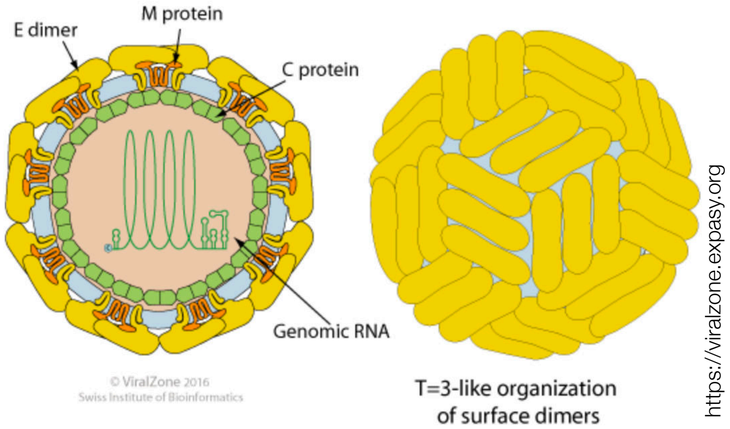


https://en.wikipedia.org

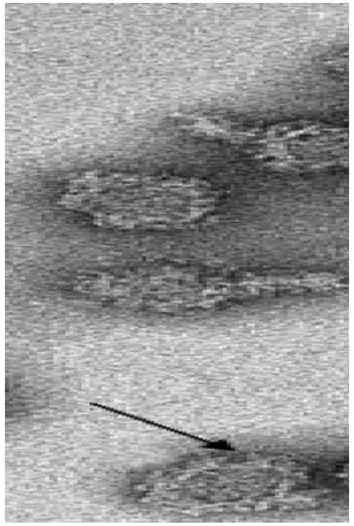


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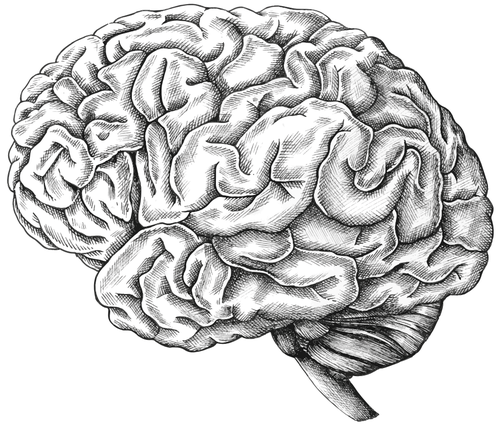
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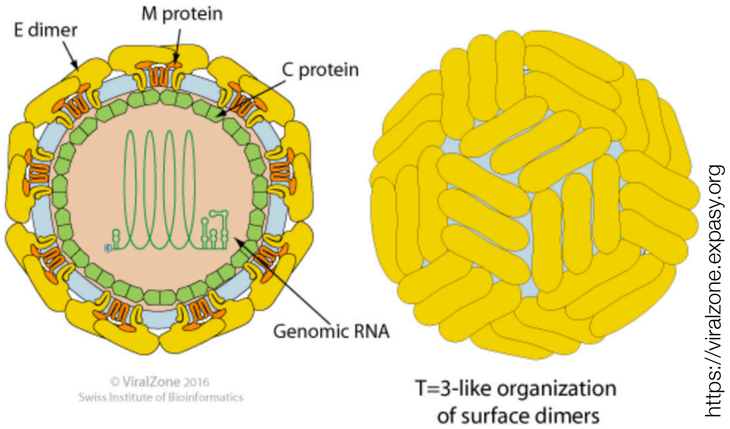
https://en.wikipedia.org



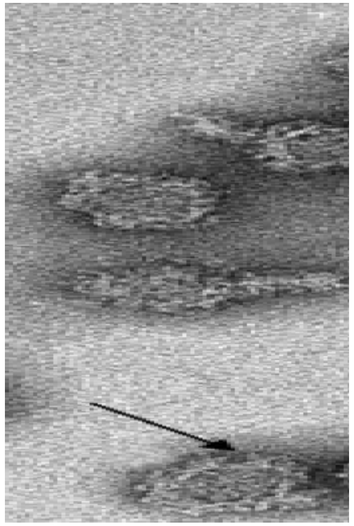
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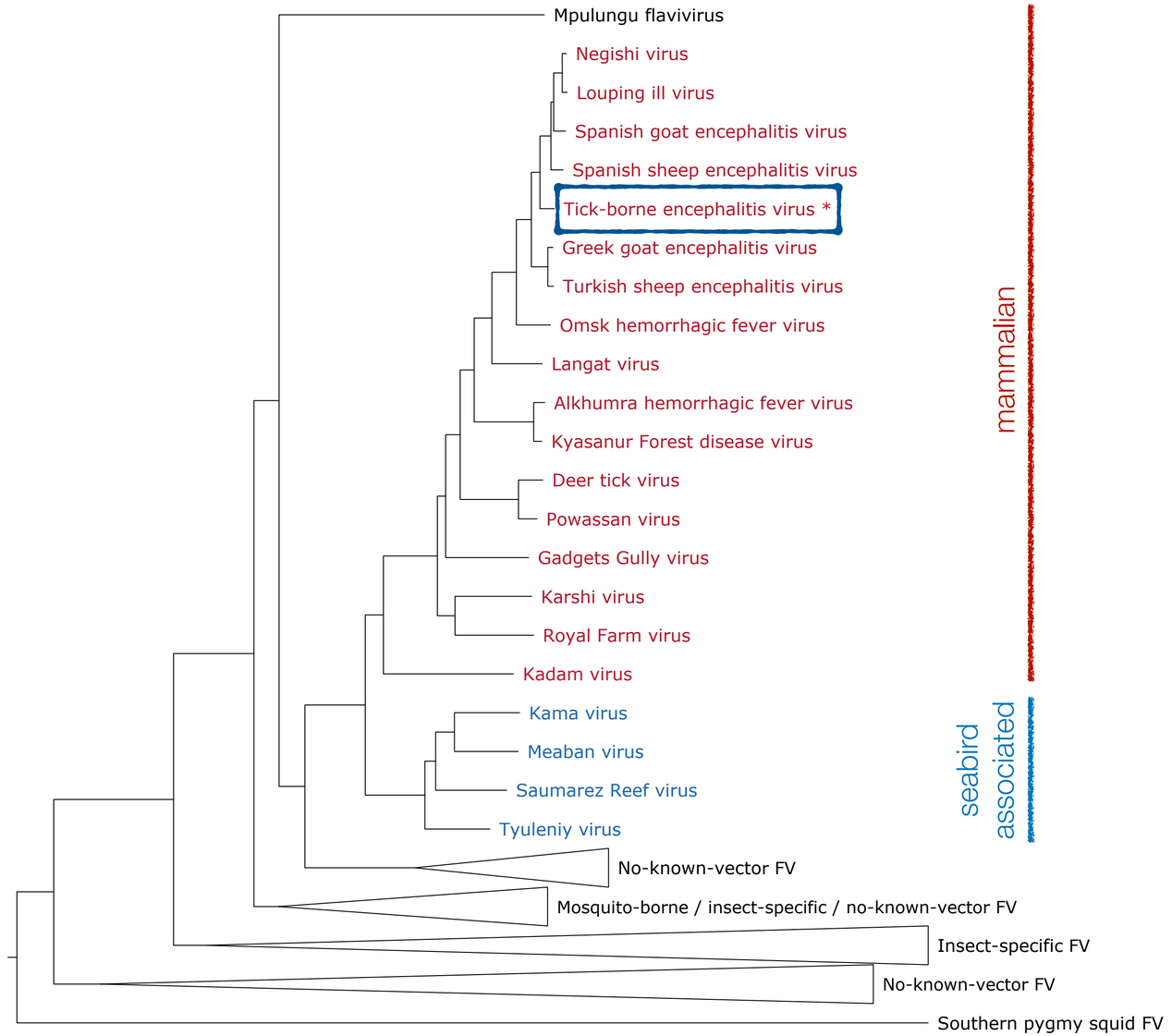


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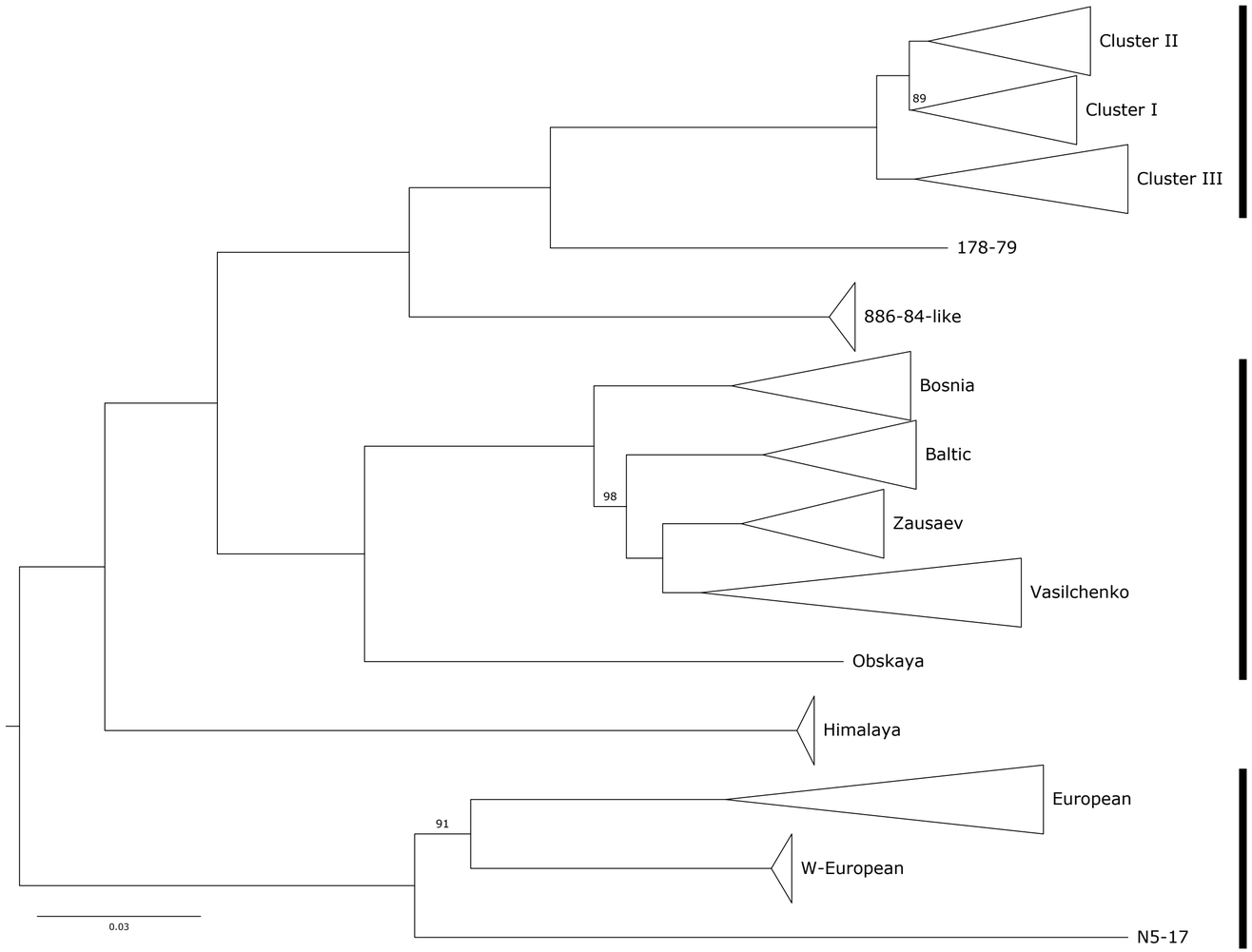
Phylogeny of tick-borne flaviviruses



Tick-borne flaviviruses

Phylogeny of tick-borne encephalitis virus

Tick-borne encephalitis virus subtyes

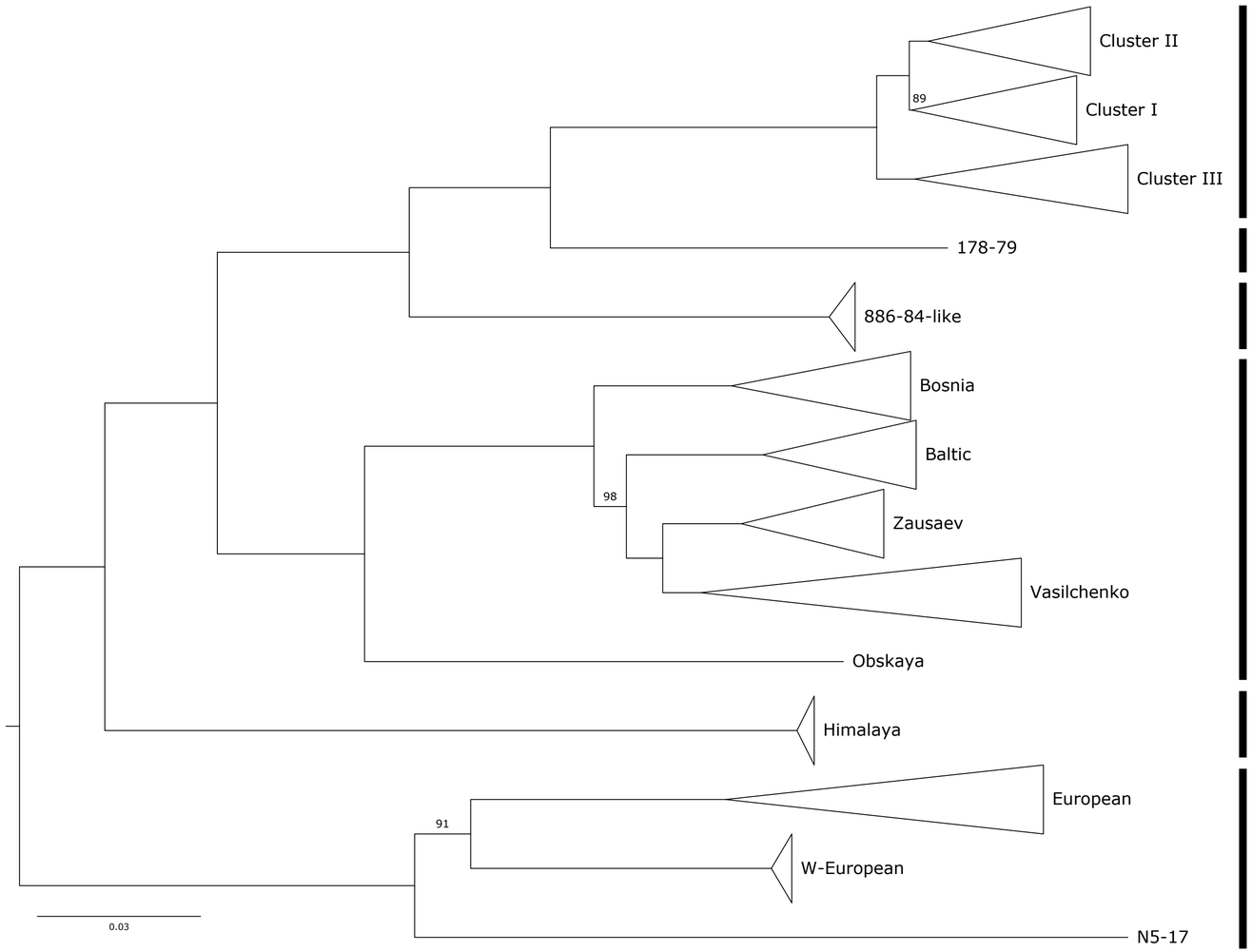


Far Eastern (TBEV-FE)

Siberian (TBEV-Sib)

European (TBEV-Eur)

Phylogeny of tick-borne encephalitis virus



Far Eastern (TBEV-FE)

Baikalean-1 (TBEV-Bkl1)

Baikalean-2 (TBEV-Bkl2)

Siberian (TBEV-Sib)

Himalayan (TBEV-Him)

European (TBEV-Eur)

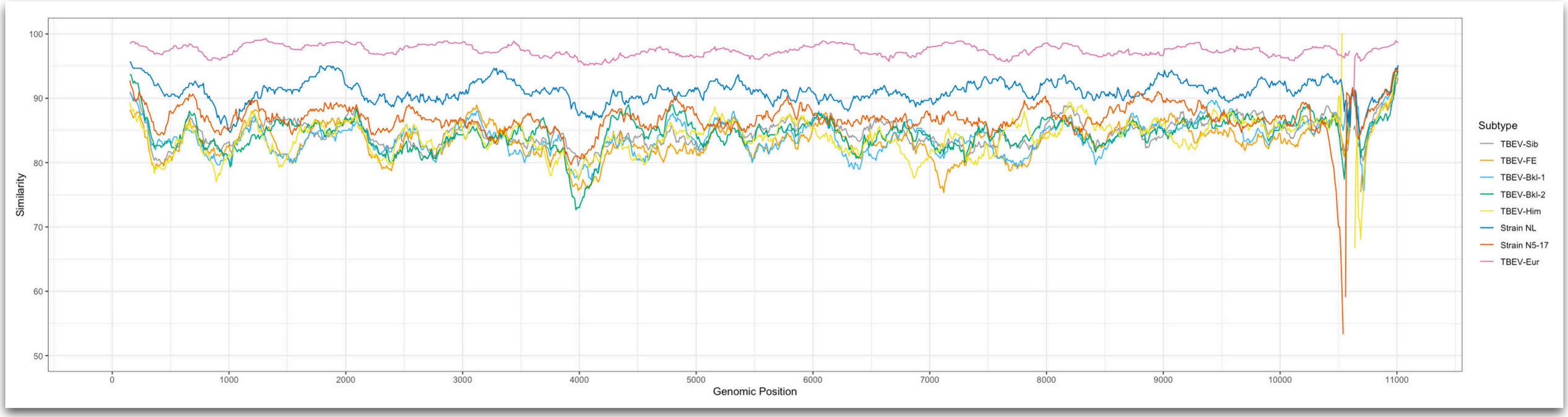
- At least6 subtypes
- Disease severity
  FE > Sib > EUR

TBEV subtype diversity

Sequence-level comparison of known TBEV subtypes vs Neudoerfl strain (300nt windows; 10nt steps)

`RNAsimplot`

technology preview



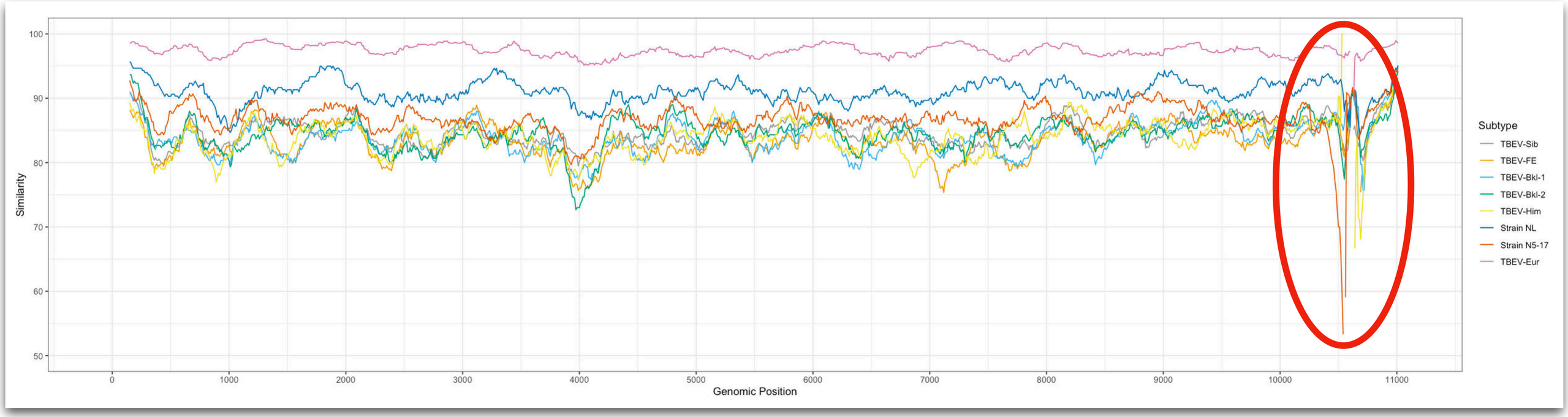
M. Riedl, I.L. Hofacker, M.T.Wolfinger (in preparation)

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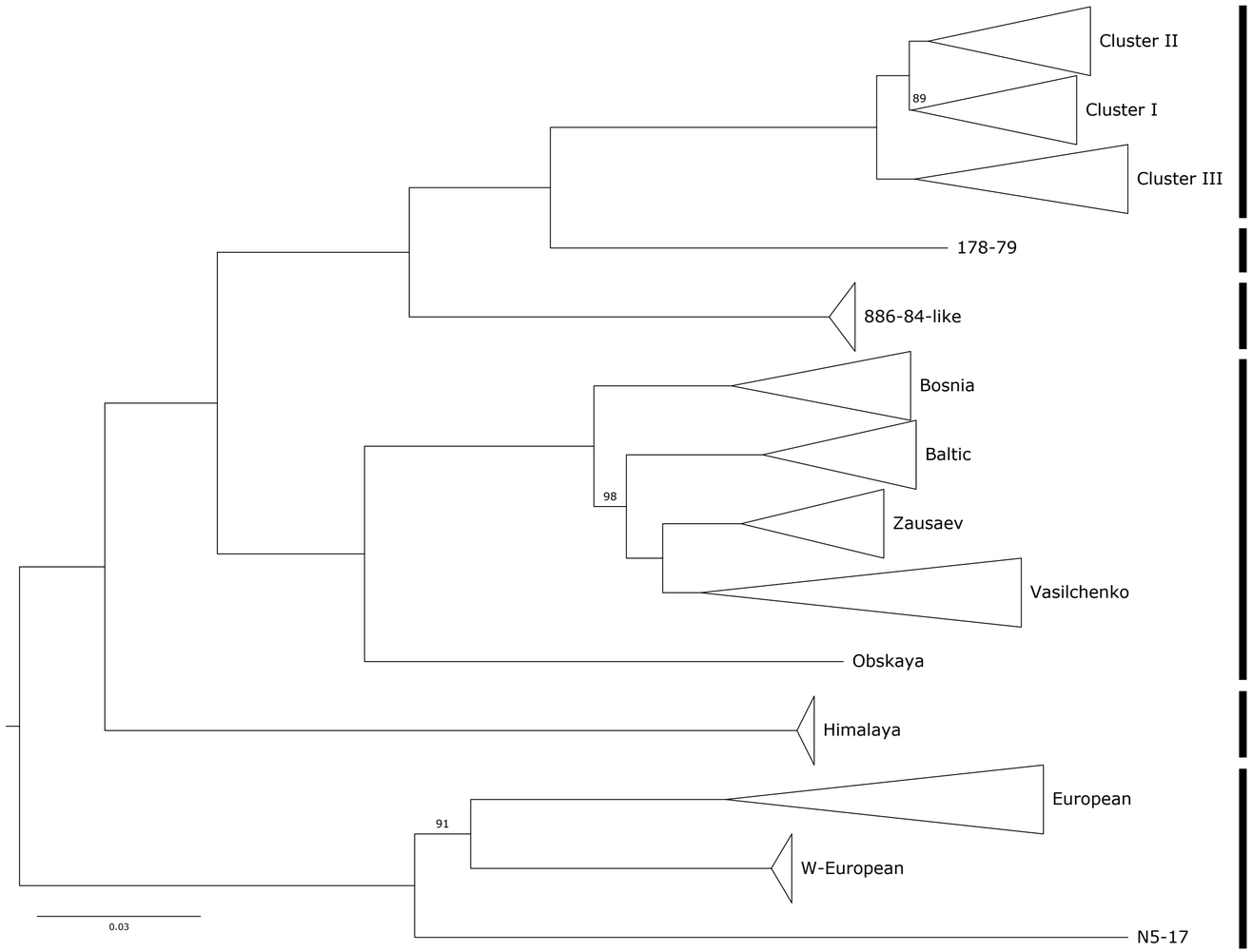
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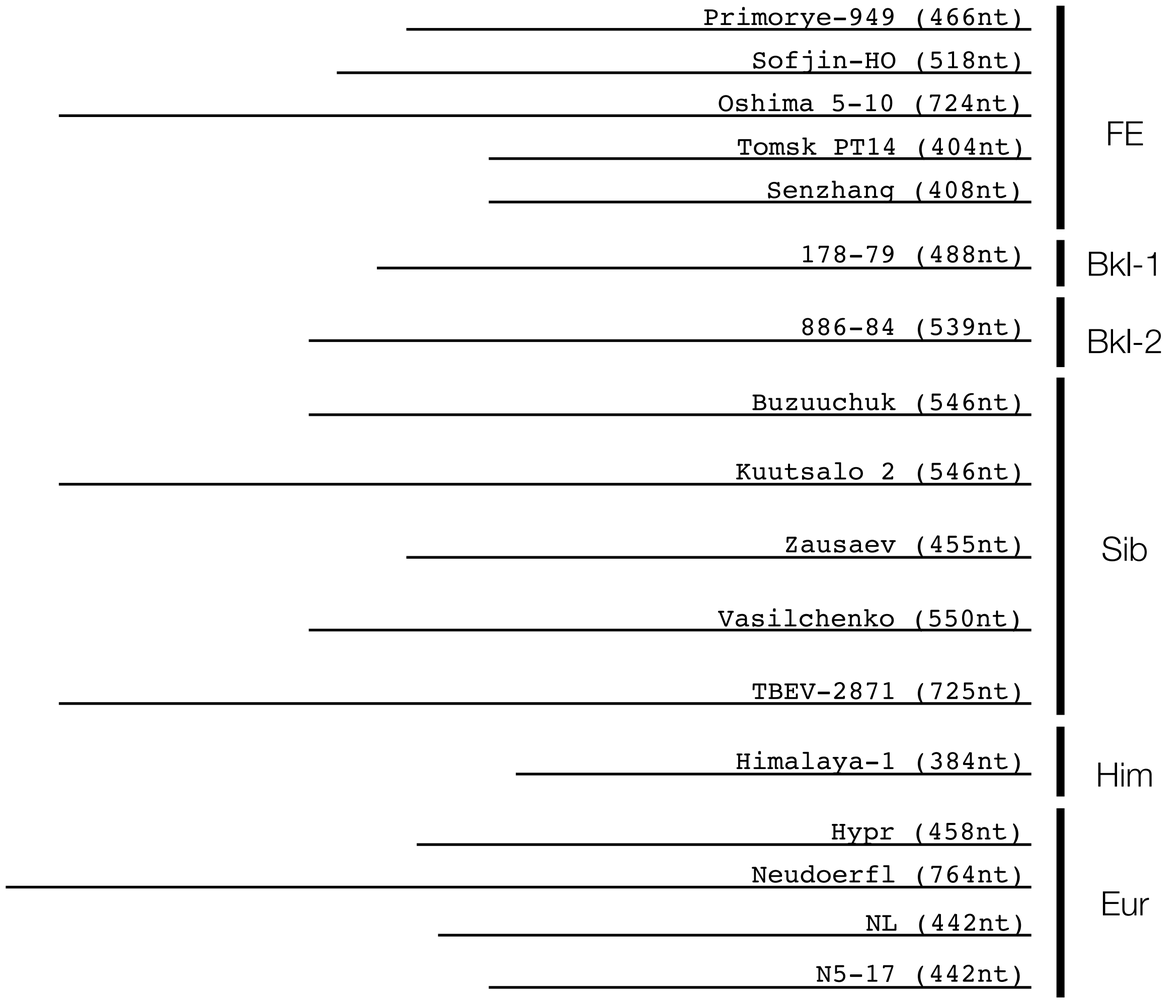


M. Riedl, I.L. Hofacker, M.T.Wolfinger (in preparation)

TBEV subtype diversity: 3’UTR

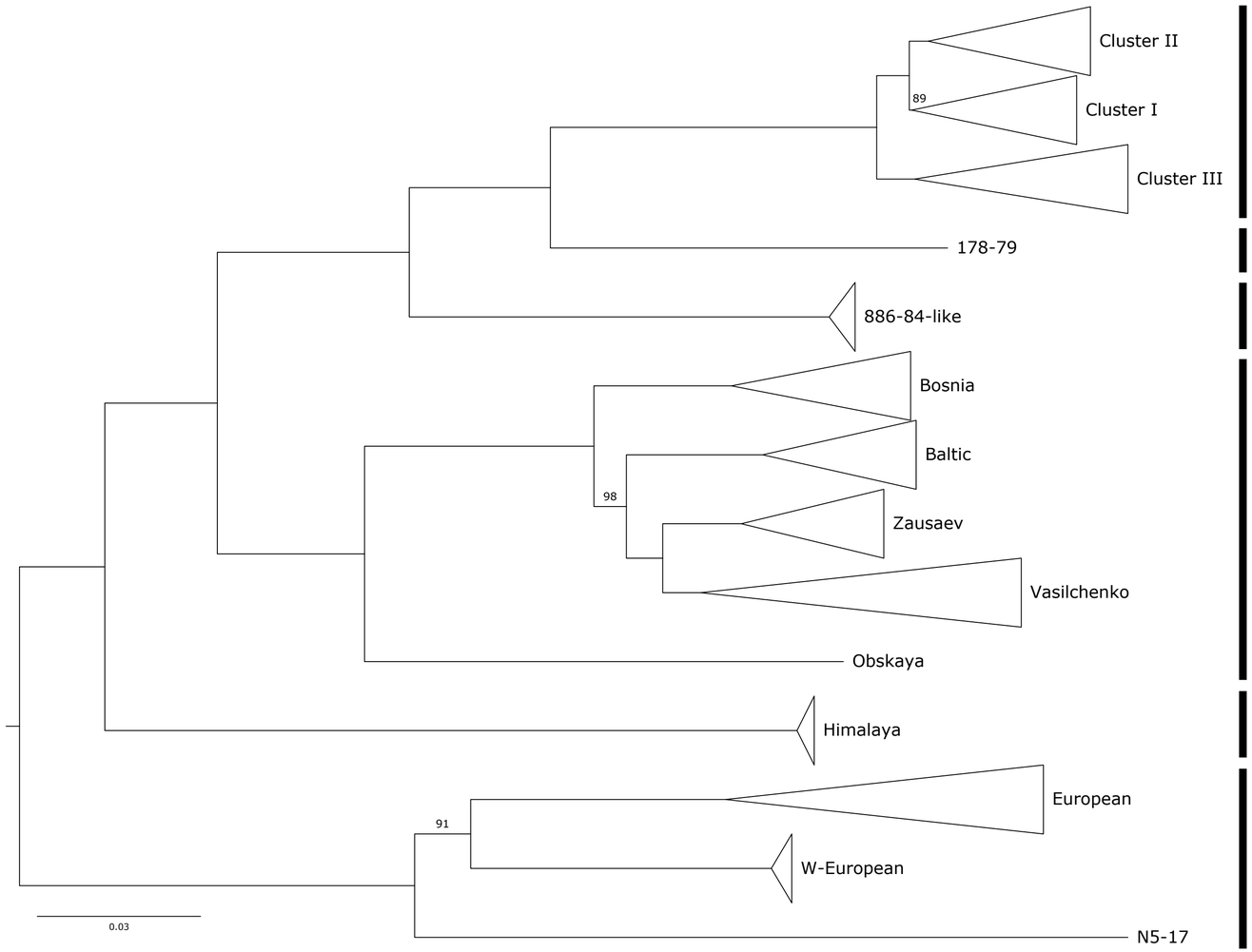
Tick-borne encephalitis virus subtyes differ in 3’UTR length

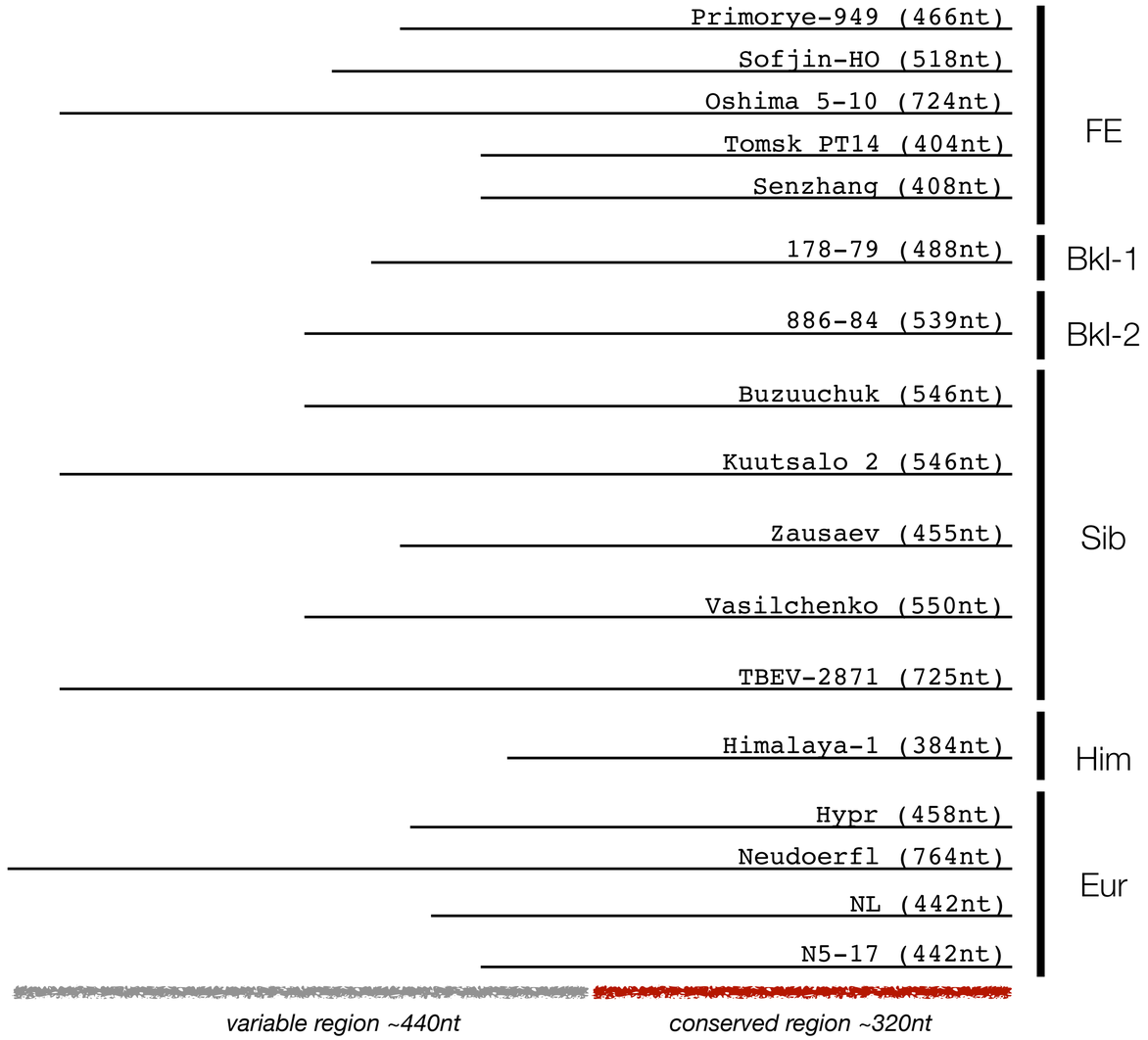




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Tick-borne encephalitis virus subtyes differ in 3’UTR length

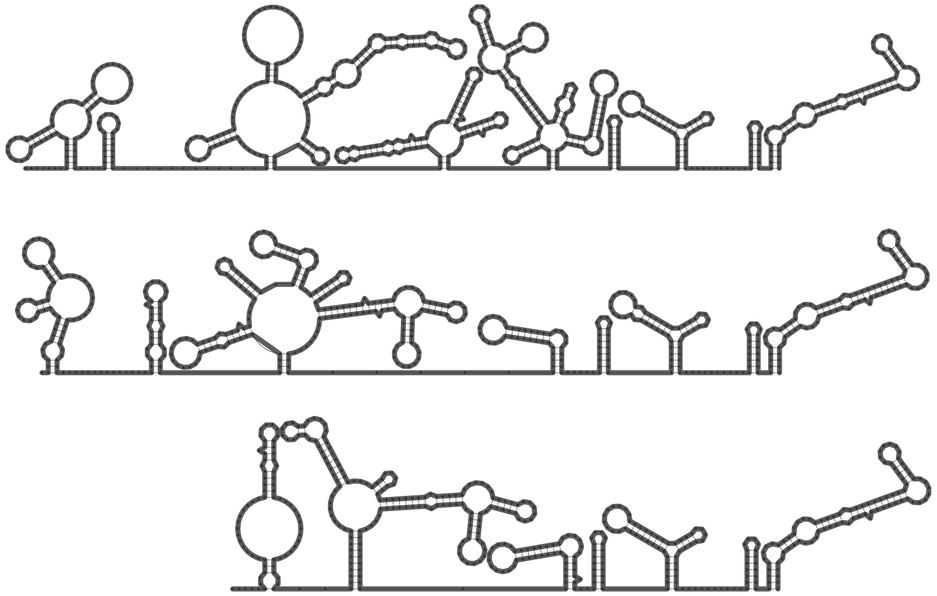


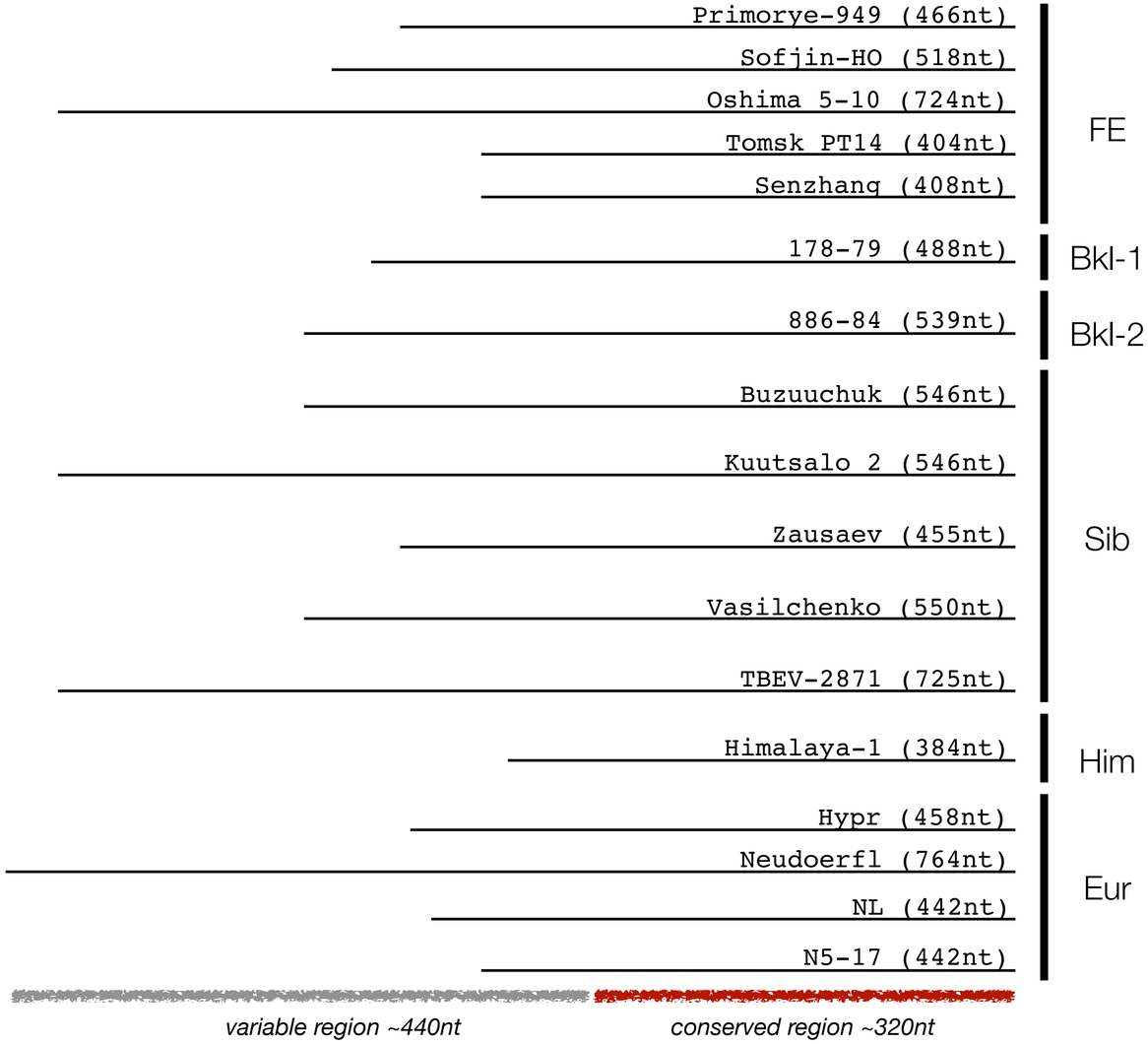


TBEV subtype diversity: 3’UTR

Tick-borne encephalitis virus subtyes differ in 3’UTR length and architecture

Single-sequence predictions

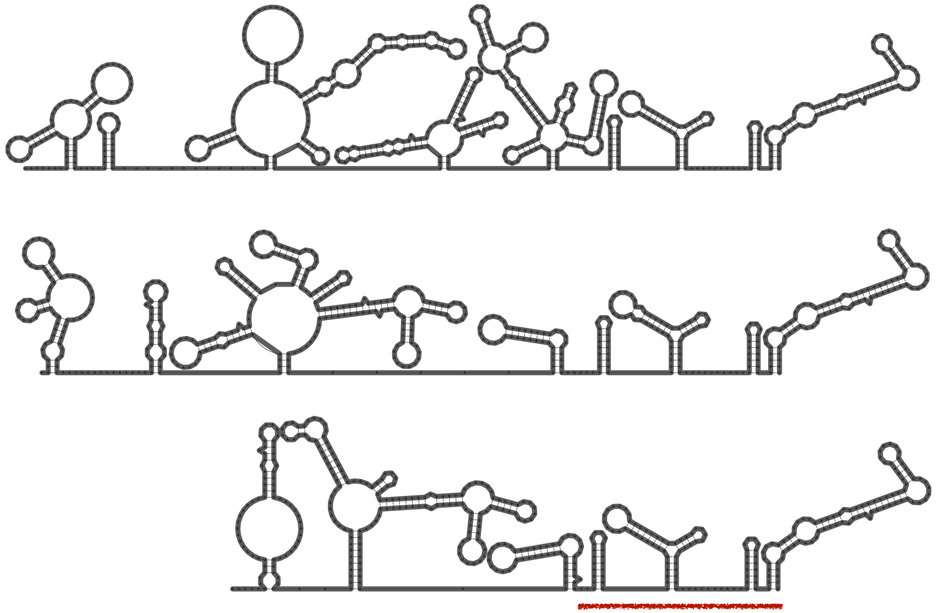




TBEV subtype diversity: 3’UTR

Tick-borne encephalitis virus subtyes differ in 3’UTR length and architecture

Single-sequence predictions



Primorye-949 (466nt)
Sofjin-HO (518nt)
Oshima 5-10 (724nt)
Tomsk PT14 (404nt)
Senzhang (408nt)
FE

178-79 (488nt)
Bkl-1

886-84 (539nt)
Bkl-2

Buzuuchuk (546nt)
Kuutsalo 2 (546nt)
Zausaev (455nt)
Vasilchenko (550nt)
TBEV-2871 (725nt)
Sib

Himalaya-1 (384nt)
Him

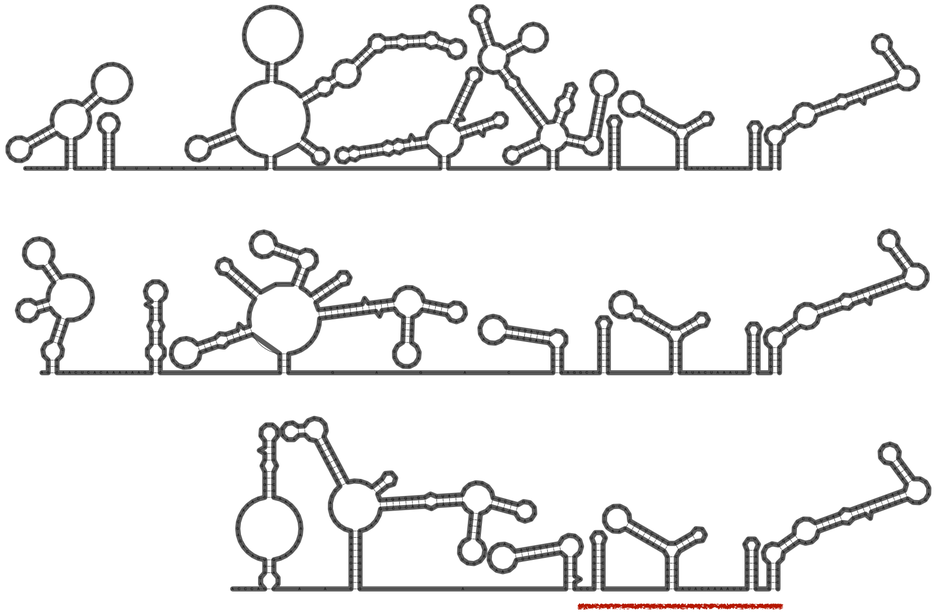
Hypr (458nt)
Neudoerfl (764nt)
NL (442nt)
N5-17 (442nt)
Eur

variable region ~440nt
conserved region ~320nt

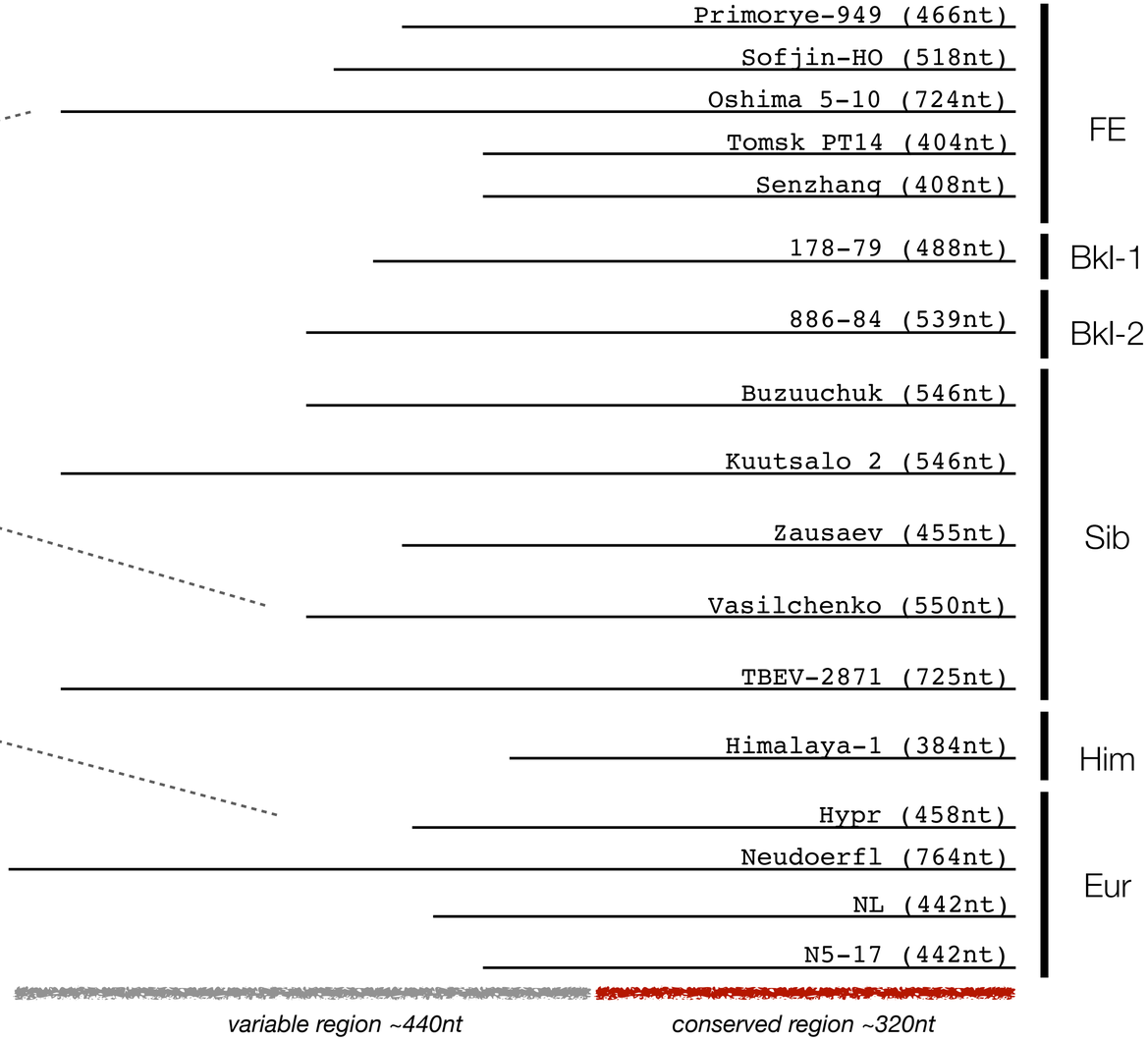
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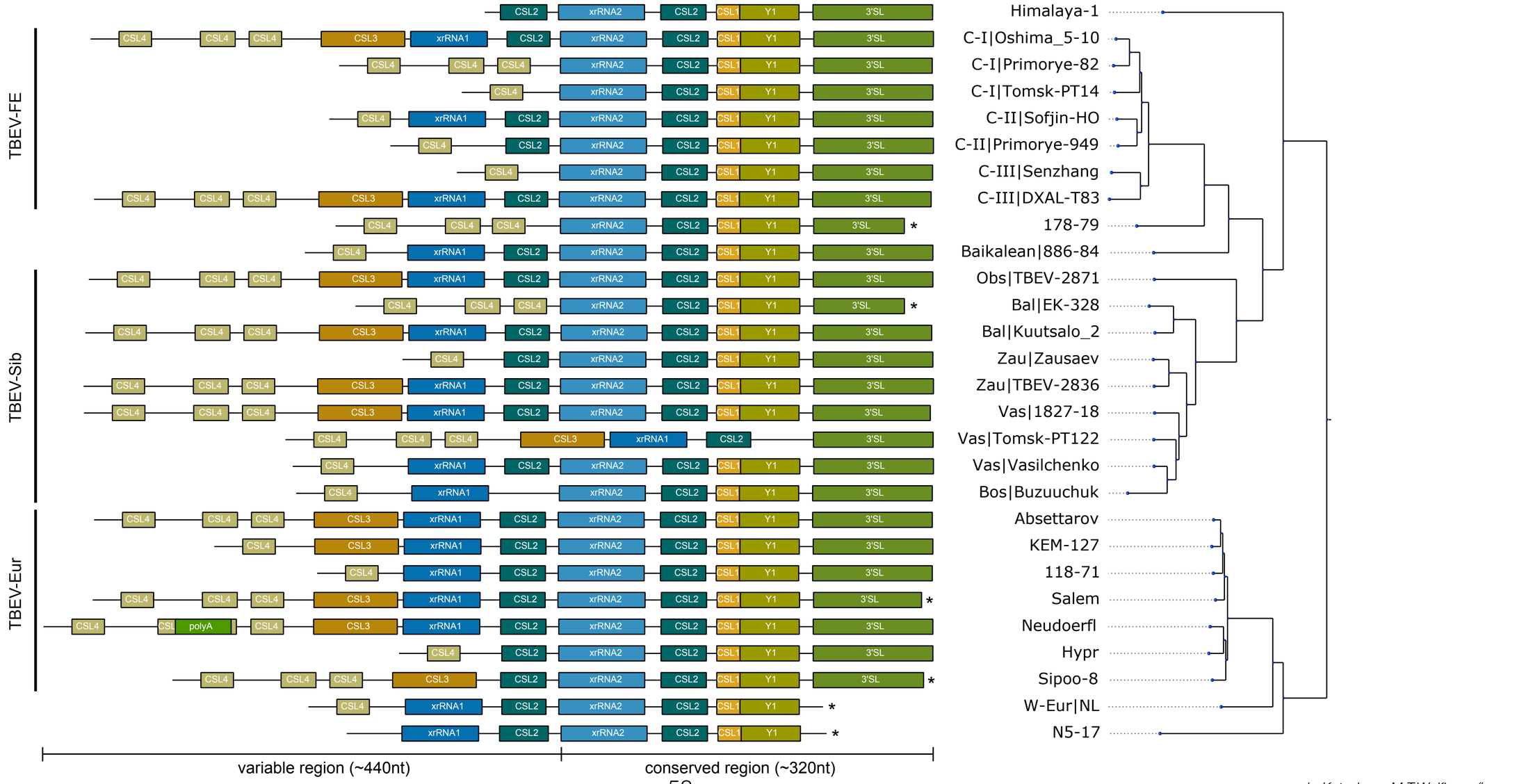
Single-sequence predictions



Prediction quality can be significantly improved by consensus structure prediction

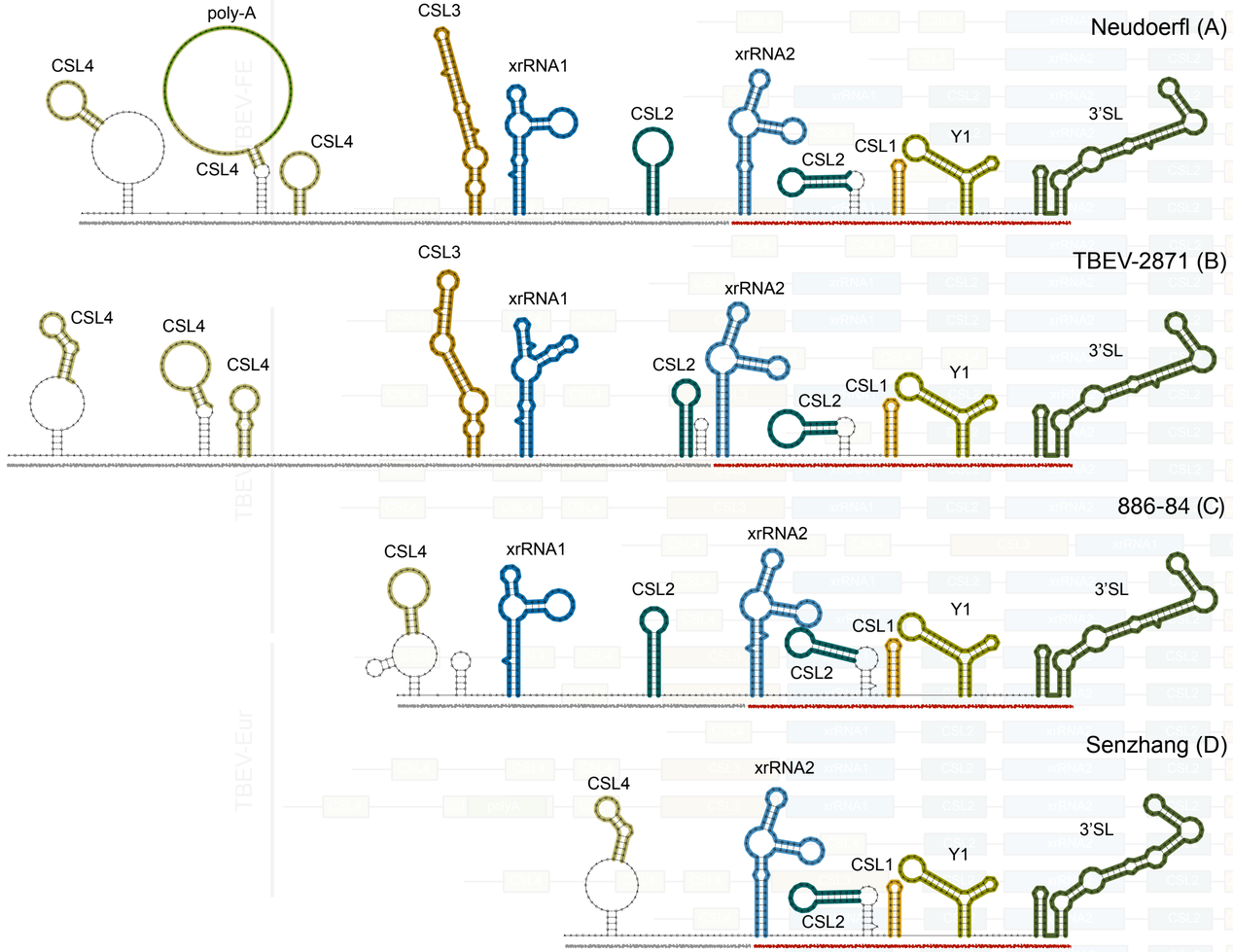


Conserved RNA elements in TBEV 3’UTRs

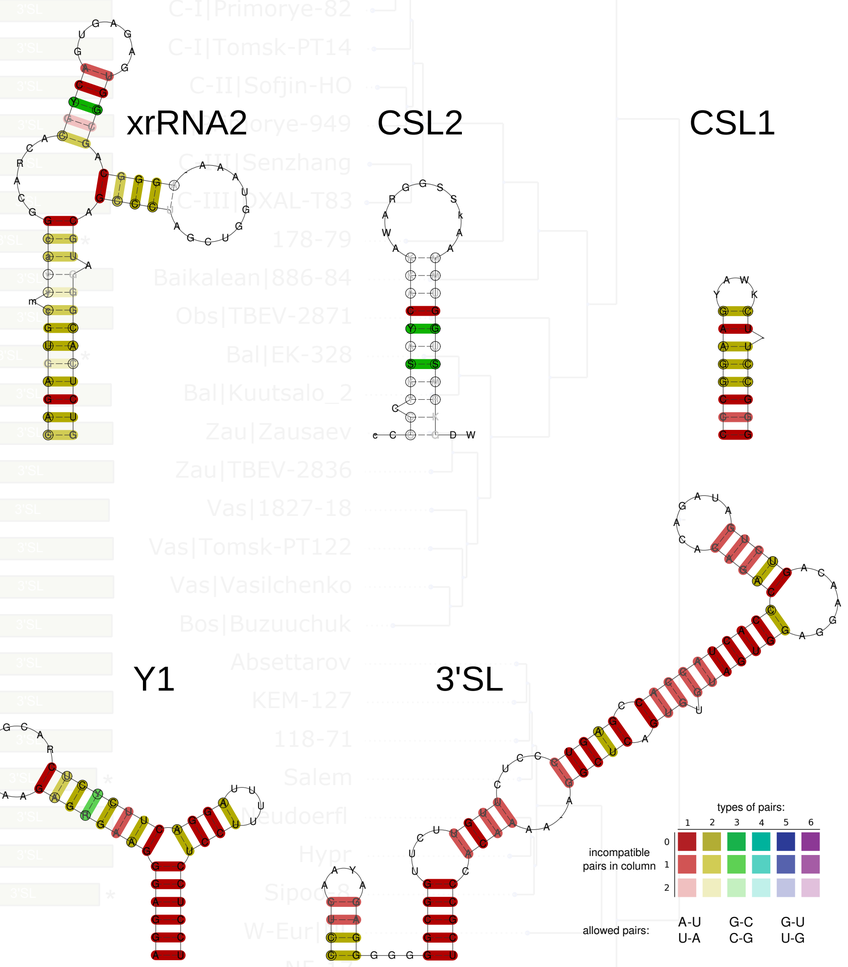


L. Kutschera, M.T.Wolfinger (in preparation)

TBEV 3'UTR: Conserved region elements

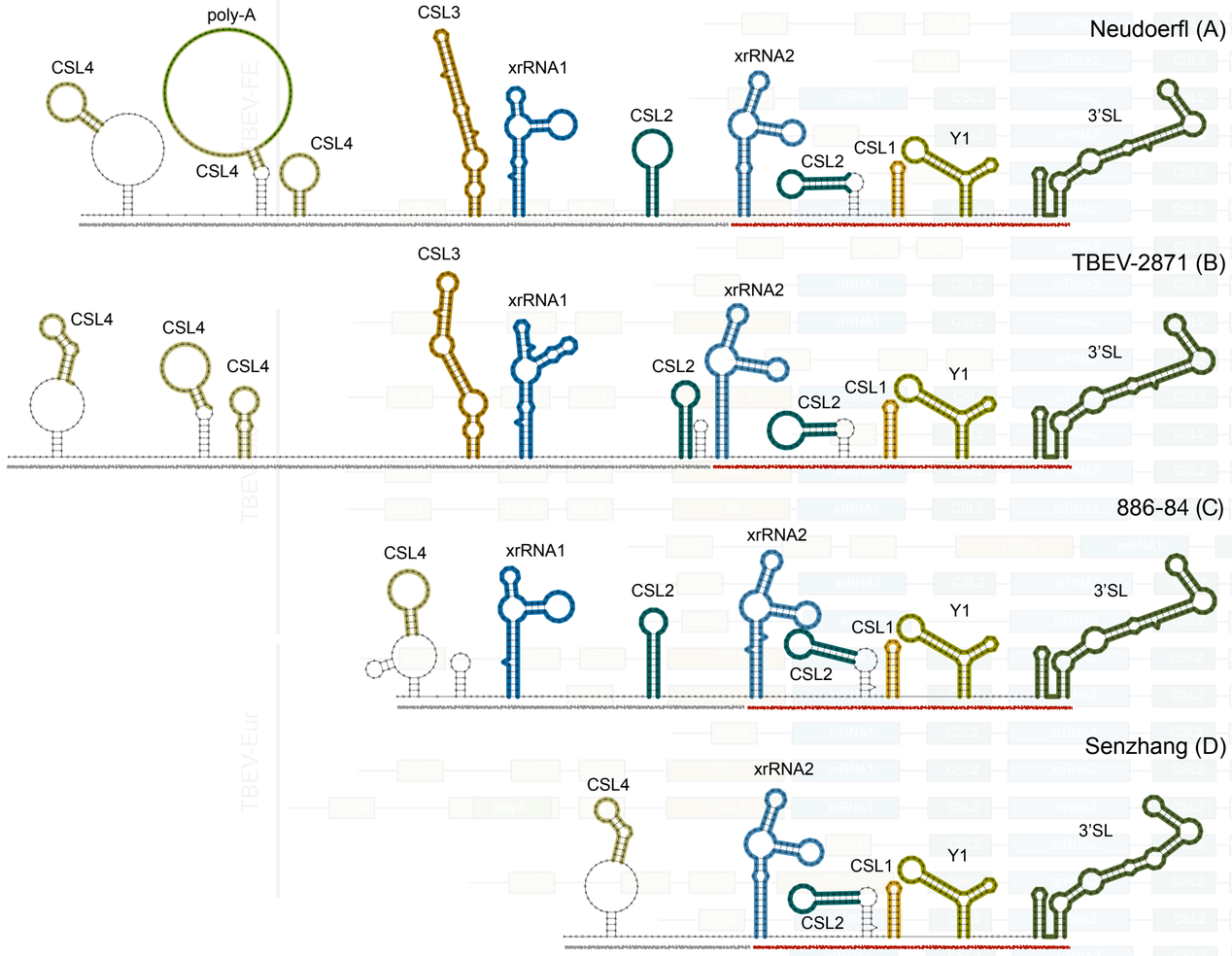


Consensus secondary structures

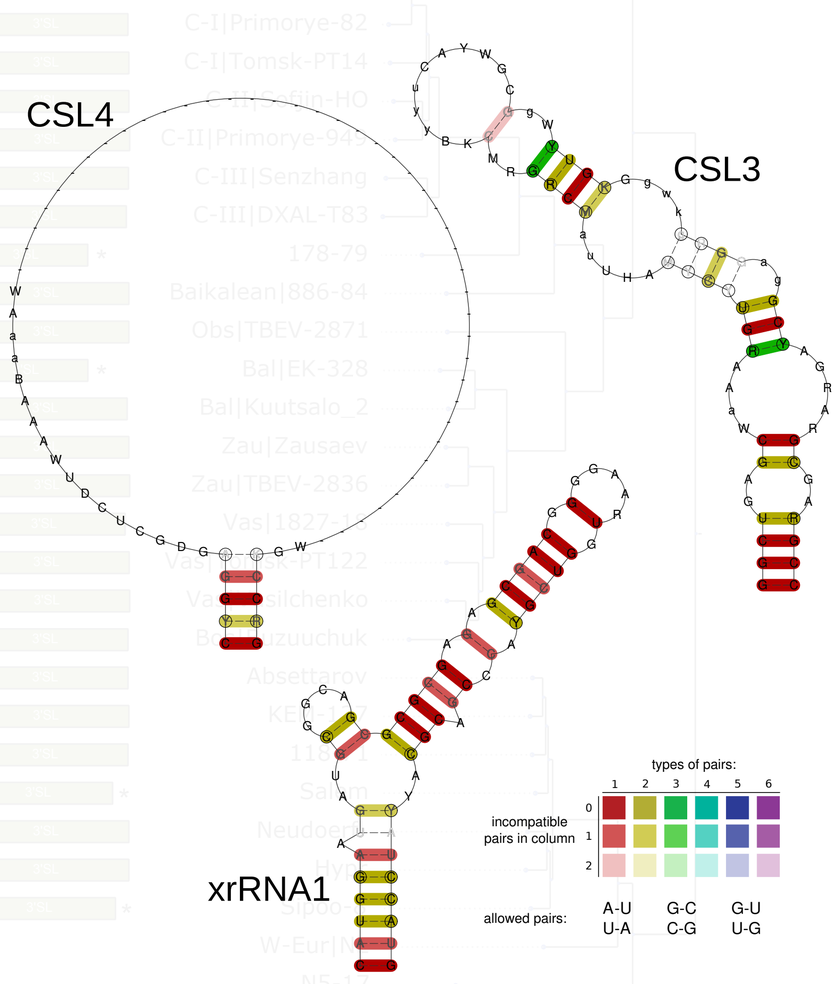


L. Kutschera, M.T.Wolfinger (in preparation)

TBEV 3’UTR: Variable region elements

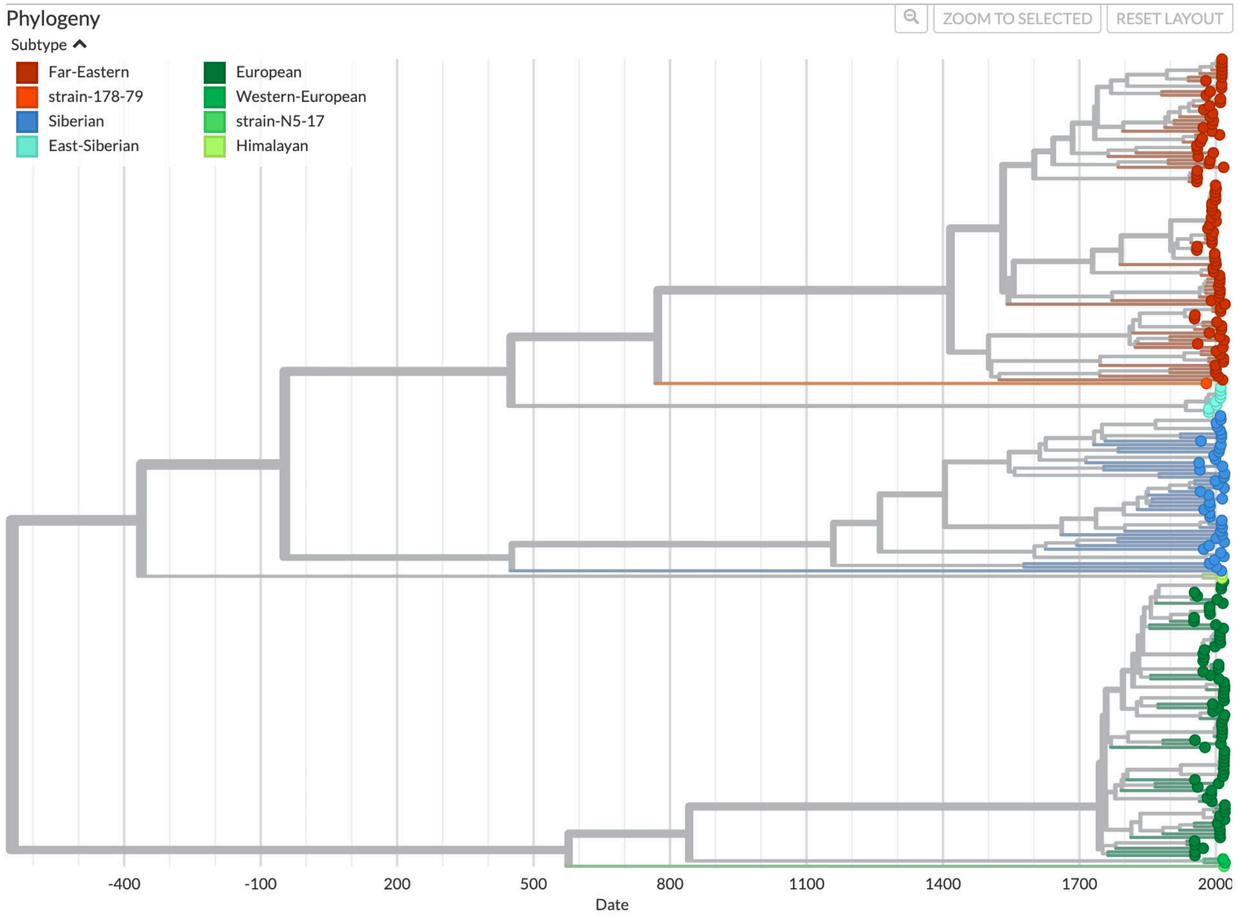


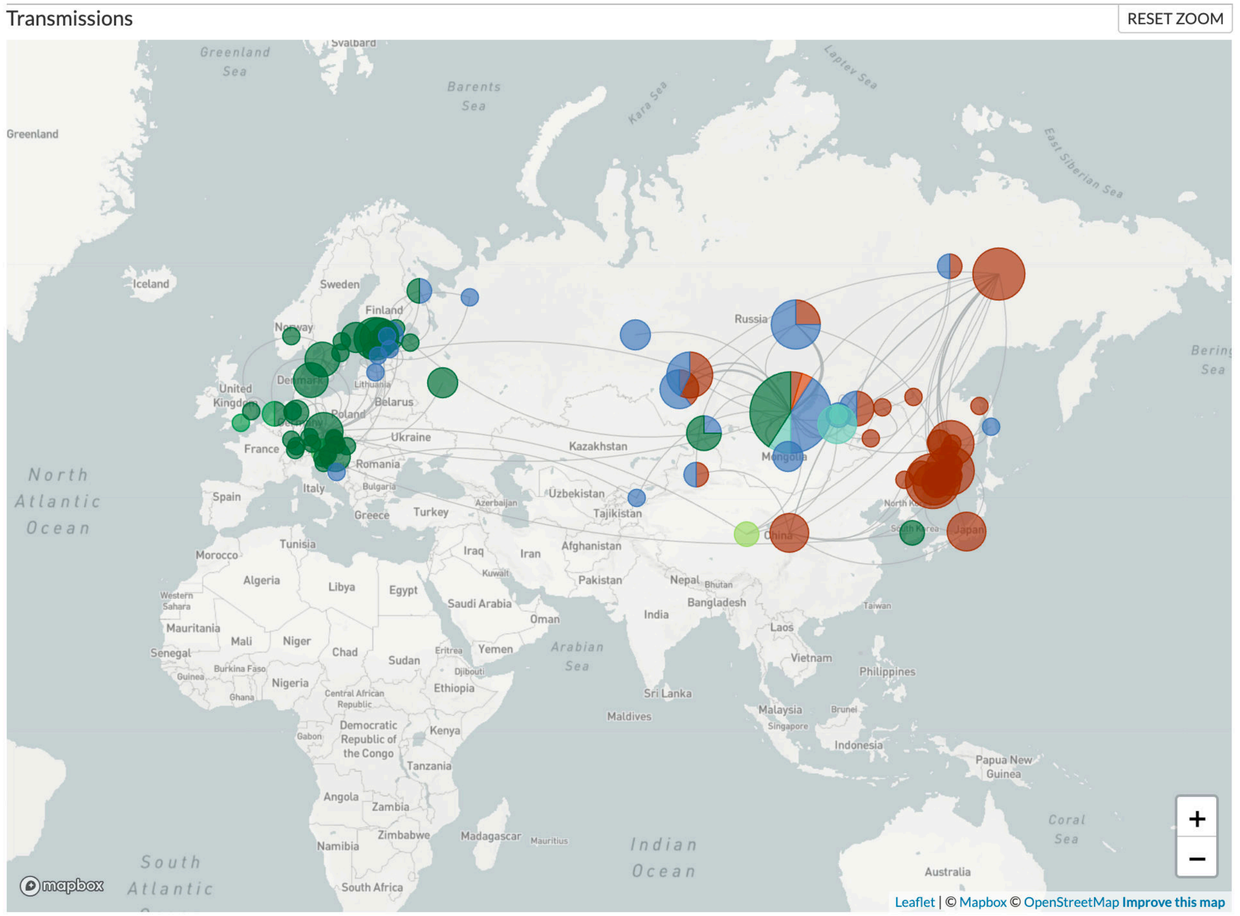
Consensus secondary structures



L. Kutschera, M.T.Wolfinger (in preparation)

TBEV molecular epidemiology





https://nextstrain.org/groups/ViennaRNA/TBEVnext

Summary

- The **`ViennaRNA`** Package provides efficient implementations for RNA structure prediction
- Structural alignments and covariance models are used to find conserved RNAs
- Pervasive conservation of functional RNA structures in TBEV 3'UTRs
- **`ViennaRNA`** Web Services are available at

http://rna.tbi.univie.ac.at/

Acknowledgements

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Lena Kutschera
Markus Riedl
Ivo Hofacker
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University of Leipzig

Mario Mörl

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